When Explanations Outlive Their Data: Faithfulness Decoupling in Graph-Attention MARL Fleet Dispatch under Telemetry Degradation

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By

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DECLARATION

I declare that this dissertation is my own original work carried out under the supervision of Dr Farhan S. Ujager and that all sources and references have been acknowledged appropriately. This dissertation has not been submitted previously for academic credit at De Montfort University, Dubai, or any other institution.

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LIST OF FIGURES

Figure 1.1 Stale data and the explanation-release decision....3
Figure 3.1 Spatial distribution of generated demand and tunnel triggers....15
Figure 3.2 Demand arrival and origin-destination distributions by split....15
Figure 3.3 Local observation graph and request-to-action mapping....17
Figure 3.4 Graph-attention policy and audit channel....18
Figure 3.5 Training, selection and held-out evaluation....21
Figure 3.6 Tunnel-triggered observation-layer telemetry degradation....22
Figure 3.7 Construct-validity controls for action-linked request nodes....24
Figure 4.1 GAT and GAT-Outage training and checkpoint-selection diagnostics....33
Figure 4.2 Clean-telemetry completed journeys by training seed....34
Figure 4.3 Faithfulness perturbation controls by checkpoint....36
Figure 4.4 Attention aggregation sensitivity....37
Figure 4.5 Attention query-row sensitivity....38
Figure 4.6 Top-k overlap and DEF resolution....39
Figure 4.7 Outage-duration sweep by policy and training seed....40
Figure 4.8 Paired attention and faithfulness shift by training seed....42
Figure 4.9 Faithfulness analysis by action type....43
Figure 4.10 Tunnel-triggered and random telemetry-loss sensitivity....44
Figure 4.11 WAMSN-DEF relationship and H4 result....44
Figure 5.1 Evidence interpretation summary....49
Figure 5.2 Freshness-aware explanation audit workflow....52

LIST OF TABLES

Table 2.1 Critical comparison of eight closely related explanation studies........................12
Table 3.1 Observation graph node types and roles..16
Table 3.2 Policy conditions used in the experiment...18
Table 3.3 Validation-selected checkpoint indices (zero-based)..19
Table 3.4 Training and optimization settings...19
Table 3.5 Checkpoint-acceptance criteria..20
Table 3.6 Explanation measures and roles...23
Table 3.7 Held-out evaluation structure and episode counts..25
Table 3.8 Telemetry conditions used in held-out evaluation..25
Table 3.9 Hypothesis map and analysis roles...27
Table 3.10 Implementation components and design rationale..30
Table 4.1 Policy capability on held-out demand..34
Table 4.2 Protocol correction and numeric-precision audit..35
Table 4.3 Decision-relevance controls under clean and degraded telemetry......................35
Table 4.4 Valid node-count distribution in scored graphs...38
Table 4.5 Exposure-conditioned paired attention and DEF shifts......................................41
Table 4.6 Action composition and paired DEF by action type...42
Table 4.7 Tunnel and random-trigger sensitivity summary..43
Table 4.8 Hypothesis outcomes across training seeds..45
Table 4.9 Freshness-aware explanation audit decisions...46
Table 4.10 Exploratory no-op margin-DEF by checkpoint and condition..........................47
Table 5.1 Freshness-aware explanation audit framework...51
Table 5.2 Audit framework inputs and outputs..52
Table C.1 Rerun per-checkpoint hypothesis statistics..66

LIST OF ABBREVIATIONS

Abbreviation	Full form
AoI	Age of Information
BMG-Q	Localized Bipartite Match Graph Attention Q-learning
CI	Confidence Interval
CTDE	Centralized Training with Decentralized Execution
DEF	Decision-level Explanation Faithfulness
GAT	Graph Attention Network
GNN	Graph Neural Network
GxI	Gradient x Input
LOO	Leave-One-Out
MADDPG	Multi-Agent Deep Deterministic Policy Gradient
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
MLP	Multilayer Perceptron
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
SUMO	Simulation of Urban Mobility
WAMSN	Weighted Attention Mass on Stale Nodes
XRL	Explainable Reinforcement Learning

ABSTRACT

Graph-attention multi-agent reinforcement learning (MARL) can expose attention weights as explanations of fleet-dispatch decisions. However, an attention map may remain visible after its supporting vehicle telemetry becomes stale. This dissertation asks whether raw graph-attention weights can be trusted as explanations under clean and degraded telemetry.

The experiment simulates 20 taxis in Simulation of Urban Mobility (SUMO). Tunnel entry triggers an observation-layer outage that freezes an affected taxi's last valid position and speed for 10, 20, 30, or 60 seconds while SUMO retains the current traffic state. This produces paired clean and degraded observations for the same decision. Graph-attention network (GAT) policies trained on clean data or with 30-second outages are evaluated across five independent training seeds. Decision-level explanation faithfulness (DEF) compares attention-ranked nodes with type-matched, action-protected random controls. Weighted attention mass on stale nodes (WAMSN) separately measures attention-weighted vehicle-information age. Further checks cover leave-one-out rankings, attention extraction, action type, and random telemetry-loss triggers.

The results do not validate the tested layer- and head-averaged self-row attention channel as a reliable explanation. Under clean telemetry, pooled dispatch-action margin-DEF is below its matched-random control, while a leave-one-out positive control is above zero. Longer outages consistently increase stale-data exposure, but faithfulness does not respond consistently across checkpoints. Attention shifts toward stale nodes in seven checkpoints and away in three. Paired probability-DEF intervals are positive in seven, negative in two, and inconclusive in one. Both seed-45 checkpoints support the predicted duration- and staleness-related decline; most checkpoints do not. No-op and dispatch directions differ in nine checkpoints, and alternative attention heads or layers can reverse the stale-attention result. Therefore, the tested weights do not provide stable evidence of decision relevance.

The dissertation develops and applies a freshness-aware explanation audit framework that reports freshness and action composition separately, applies action-aware faithfulness tests, checks the attention extraction rule, and requires consistent evidence across checkpoints. Attention remains an internal diagnostic when these checks fail. Applied to the collected evidence, the framework returns WITHHOLD for both model families and all ten checkpoints, restricting the tested attention channel to diagnostic use.

Keywords: explainable reinforcement learning; graph attention; multi-agent reinforcement learning; fleet dispatch; telemetry degradation; faithfulness.

TABLE OF CONTENTS

Chapter 1: Introduction........1
1.1 Context........1
1.2 Problem statement........1
1.3 Research questions........1
1.4 Objectives........2
1.5 Main findings........2
1.6 Contributions........3
1.7 Dissertation structure........4
Chapter 2: Literature Review........5
2.1 Reinforcement learning for fleet dispatch........5
2.2 Explainability in reinforcement learning........5
2.3 Is attention an explanation?........6
2.4 Explaining graph neural networks........7
2.5 Evaluating faithfulness: methods and pitfalls........8
2.6 Age of Information and telemetry degradation........9
2.7 Deciding when an explanation can be shown........10
2.8 Research gap........10
Chapter 3: Methodology........13
3.1 Study design and proposed approach........13
3.2 Dataset selection and exploratory data analysis........13
3.2.1 Data generation and selection rationale........14
3.2.2 Spatial and temporal exploration........14
3.2.3 Policy-conditioned observations and exposure........16
3.3 Observation graph and action space........16
3.4 Policy models and training........17
3.5 Telemetry degradation........21
3.6 Explanation measures........22
3.6.1 Decision-level explanation faithfulness........22
3.6.2 Stale-node attention........23
3.7 Construct-validity audit........24
3.8 Evaluation design........25
3.8.1 Primary evaluation........25
3.8.2 Supporting analyses........26
3.9 Hypotheses and statistical analysis........27
3.9.1 Statistical procedure........27
3.9.2 Interpretation limits........28
3.10 Audit decision implementation........28
3.11 Implementation and reproducibility........29
3.11.1 Software environment and provenance........29
3.11.2 Component interfaces and feature configuration........30
3.11.3 End-to-end algorithm........31
3.12 Ethics and data governance........32
Chapter 4: Results........33

4.1 Policy capability and training stability....33
4.2 Protocol correction and precision audit....35
4.3 Decision-relevance controls....35
4.4 Small-graph resolution....38
4.5 Telemetry manipulation and stale exposure....39
4.6 Paired clean and degraded observations....40
4.7 Analysis by action type....42
4.8 Random-trigger sensitivity analysis....43
4.9 Hypothesis results....44
4.10 Answers to the research questions....45
4.11 Explanation-release audit....46
4.12 Supplementary no-op decision relevance....46
Chapter 5: Discussion....48
5.1 Answer to the central problem....48
5.2 How to read the evidence....48
5.3 Why the result is not consistent....49
5.4 Why the construct-validity controls matter....50
5.5 Why outage training did not solve the problem....51
5.6 Proposed audit framework....51
5.6.1 Purpose and scope....51
5.6.2 Inputs, outputs and use....52
5.6.3 Application to this study....53
5.7 Limitations....54
5.8 Future work....55
Chapter 6: Conclusion....56
List of Publications....58
References....59
Appendices....65

Chapter 1: Introduction

1.1 Context

Fleet dispatch is relational. A taxi's useful action depends on nearby vehicles, open passenger requests, road conditions, and competition for the same demand. Graph neural networks (GNNs) represent these relationships directly, and graph attention networks (GATs) add learned weights over neighboring nodes (Scarselli et al., 2009; Veličković et al., 2018). Because the weights are easy to display, they are often read as an explanation of which vehicles or requests influenced a decision.

Interpreting these weights requires care. Studies in NLP and other attention-based models have found that attention weights can be weakly related to the evidence that changes a model's output (Jain and Wallace, 2019; Serrano and Smith, 2019; Liu et al., 2022). Wiegreffe and Pinter (2019) caution that such findings do not rule out all attention-based explanations. The issue becomes more serious in an operational system because the input itself may no longer describe the current world.

1.2 Problem statement

Vehicle telemetry can be interrupted by tunnels and other signal-loss areas. During an interruption, a dispatch platform may keep the last received position and speed. The reading remains available, but its Age of Information (AoI) grows (Kaul, Yates and Gruteser, 2012). An attention map can therefore look complete even though some nodes represent old information. The interface reveals where the model placed attention, but it does not reveal whether that information is fresh or whether removing it would change the decision.

This makes the explanation difficult to trust. An operator may read a complete attention map as a trustworthy reason even when its evidence is stale, and stable dispatch performance cannot show whether the explanation is valid. This dissertation tests the reliability of raw graph-attention explanations under clean and degraded telemetry, with dispatch performance providing context for interpreting the explanation results.

This dissertation uses *faithfulness decoupling* to describe a situation in which an explanation still appears credible after its supporting data have become stale, even though the displayed attention weights may not reliably identify the evidence that influenced the decision. One possible pattern is declining faithfulness while dispatch performance remains stable. The core question is whether attention provides reliable evidence under the tested conditions.

The term identifies the risk being investigated, rather than a result assumed in advance. Showing that degradation reduces faithfulness would require a valid clean faithfulness baseline and a lower score in the paired degraded observation. The rerun finds declines in some checkpoints but no consistent response across seeds. Its contribution includes identifying why the declared attention channel fails the release audit and distinguishing that model-level decision from checkpoint-specific changes under degradation.

1.3 Research questions

The central research question is:

Can raw graph-attention weights be trusted as explanations of fleet-dispatch decisions when vehicle telemetry becomes stale?

Four research questions guide the evaluation:

1. **RQ1:** Under clean telemetry, do raw graph-attention weights identify decision-relevant nodes more reliably than type-matched random controls?
2. **RQ2:** Does tunnel-triggered telemetry degradation change the attention assigned to stale vehicle nodes relative to the paired clean observation?
3. **RQ3:** As outage duration increases, are changes in stale-node attention and decision-level faithfulness consistent across independently trained checkpoints?
4. **RQ4:** In the tested configurations, does degradation-aware training produce more consistent attention and faithfulness responses under telemetry degradation than clean training?

All four questions depend on a valid faithfulness measure. The evaluation therefore distinguishes the removal of information from the removal of an available action, which can otherwise distort the measured effect.

1.4 Objectives

The measurable objectives are to:

1. develop a reproducible Simulation of Urban Mobility (SUMO) fleet-dispatch benchmark that creates paired clean and degraded observations from the same simulated state and uses held-out test demand;
2. evaluate whether raw graph-attention weights identify decision-relevant nodes under clean telemetry using action-protected, type-matched controls and a leave-one-out (LOO) positive control;
3. quantify the paired change in attention assigned to stale vehicle nodes under tunnel-triggered outages and test its sensitivity to randomly triggered telemetry loss;
4. assess whether stale-node attention and decision-level faithfulness responses are consistent across outage durations, independently trained checkpoints, action groups, and attention-extraction choices;
5. compare the matched clean-trained and degradation-aware configurations to assess whether the tested degradation-aware configuration gives more consistent attention and faithfulness responses;
6. implement and apply a freshness-aware explanation audit framework that converts the collected evidence into an ELIGIBLE, WITHHOLD, or INCOMPLETE explanation-release decision.

Objectives 1-2 provide the benchmark and clean controls for RQ1; objective 3 addresses the paired telemetry comparison in RQ2; objective 4 tests repeatability and sensitivity for RQ3; and objective 5 addresses RQ4. Objective 6 uses these findings to reach the audit decision. The answers and release outcomes are presented in Sections 4.10-4.11.

1.5 Main findings

The tested self-row attention channel lacks consistent evidence of decision relevance and stability. Figure 1.1 summarizes the practical consequence: an attention map can remain visible as telemetry becomes stale, even when the evidence is insufficient to present it as an explanation.

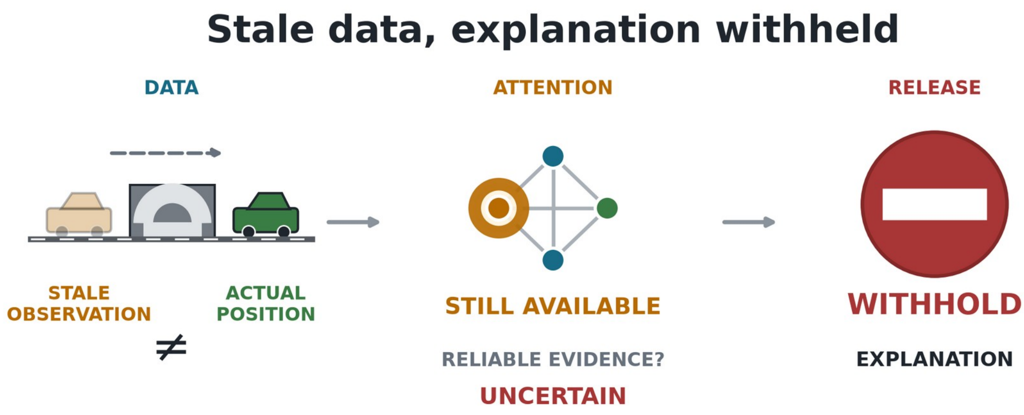

Figure 1.1. Main finding. A tunnel-triggered outage leaves the policy with a stale observation. Attention remains available, but the audit withholds it because decision relevance and extraction consistency are not established. Freshness visibility passes as a reporting check; it does not certify fresh data.

Under clean telemetry, the tested averaged self-row attention has dispatch margin-DEF below its type-matched random control for both model families. A leave-one-out control produces a positive response, showing that the evaluator can detect a more decision-relevant ranking even though raw attention does not provide one.

Longer outages consistently increase the measured exposure to stale vehicle information. However, the paired response differs across trained policies: seven checkpoints shift attention toward stale nodes and three shift away. Probability-DEF intervals are positive for seven checkpoints, negative for both seed-45 checkpoints, and inconclusive for GAT-Outage seed 46. The study finds degradation-related loss of faithfulness in some checkpoints, but no consistent effect across all seeds.

The interpretation also depends on how attention is extracted and whether the selected action is dispatch or no-op. Nine checkpoints have opposite paired DEF directions for these action groups. The selected-request-row control is positive for GAT-Outage but negative for GAT, although the default self-row control is negative for both model families. Outage training does not remove checkpoint variation.

The practical conclusion is that data freshness and explanation faithfulness must be checked separately for each trained policy. In the present experiment, the policy may still produce an action, but its raw attention weights should be withheld as an explanation.

1.6 Contributions

The dissertation makes four main contributions:

1. a reproducible paired clean/degraded benchmark that creates two observations from the same SUMO state, separates physical ground truth from policy input, and distinguishes the tunnel trigger from the observation-layer outage;

2. an action-aware, type-matched evaluation protocol that reduces the deletion bias caused by request nodes representing both information and available actions, checked through LOO, overlap, and action-type diagnostics;
3. evidence across independently trained checkpoints showing that stale-data exposure increases consistently with outage duration, while direct paired attention and DEF shifts vary across checkpoints and attention-extraction choices, with the pattern also tested under random telemetry loss;
4. an implemented freshness-aware explanation audit framework that combines freshness, faithfulness, action composition, extraction sensitivity, and checkpoint-consistency evidence to return an ELIGIBLE, WITHHOLD, or INCOMPLETE explanation-release decision.

1.7 Dissertation structure

Chapter 1: Introduction. Defines the research problem, research questions, objectives, main findings, and contributions.

Chapter 2: Literature Review. Reviews MARL fleet dispatch, graph attention, explanation faithfulness, graph explainability, and telemetry freshness.

Chapter 3: Methodology. Describes dataset selection and EDA, the SUMO environment, policy models, observation-layer degradation, faithfulness measures, evaluation design, implementation details, and audit rules.

Chapter 4: Results. Reports policy capability, construct-validity checks, faithfulness results, stale-data exposure, sensitivity analyses, and audit decisions.

Chapter 5: Discussion. Interprets the findings, explains their limitations, and presents the freshness-aware explanation audit framework.

Chapter 6: Conclusion. Answers the research questions and summarizes the practical implications for trustworthy fleet-dispatch explanations.

Chapter 2: Literature Review

This chapter reviews research on graph-based fleet dispatch, attention-based explanations, and explanation faithfulness under changing observations. It draws on foundational studies to establish the models and evaluation concepts, alongside recent work on graph attribution, delayed observations, and multi-agent communication. Particular attention is given to how these studies assess decision relevance and whether their assumptions remain valid when vehicle telemetry becomes stale.

2.1 Reinforcement learning for fleet dispatch

Fleet dispatch links vehicles, passenger requests, and a road network over time. A decision for one taxi changes the demand available to others, so a collection of independent single-agent policies can create competition or duplicated assignments. Lin et al. (2018) model city-scale fleet management as a cooperative multi-agent reinforcement-learning (MARL) problem. Qin, Zhu and Ye (2022) show that reinforcement learning (RL) is used for matching, repositioning, pricing, and route decisions, often under changing supply and demand.

MARL offers several ways to represent coordination. Multi-Agent Deep Deterministic Policy Gradient (MADDPG) learns decentralized actors with centralized critics (Lowe et al., 2017). QMIX factorizes a team value into agent values while preserving a monotonic relation to the joint action (Rashid et al., 2020). Multi-Agent Proximal Policy Optimization (MAPPO) uses the more familiar Proximal Policy Optimization (PPO) objective with centralized training and decentralized execution; its empirical stability makes it a practical baseline for cooperative tasks (Yu et al., 2022). MAPPO provides a stable training basis for examining the explanation attached to each taxi's local decision.

Graphs provide a natural representation of the local dispatch problem. Nodes can represent the acting taxi, nearby taxis, and open requests; edges permit their information to be combined (Scarselli et al., 2009). A graph attention network (GAT) assigns learned weights while aggregating neighboring nodes (Veličković et al., 2018). Actor-Attention-Critic uses attention in its centralized critic to select information from other agents (Iqbal and Sha, 2019). GAT-MARL is a suitable test case because it represents the relational, multi-agent structure of fleet dispatch and exposes attention weights for audit. These weights are easy to visualize, which makes them tempting to present as reasons for an action. Ease of display, however, is not evidence that the weights identify the information that caused the decision.

Recent dispatch methods continue to use relational structure. Hu, Feng and Li (2025) propose Localized Bipartite Match Graph Attention Q-learning (BMG-Q), which uses a local bipartite graph for ride-pooling dispatch. CoopRide studies cooperation across city grids (Wang et al., 2025), while DualG-MARL combines vehicle-state and task graphs (Sha et al., 2026). These studies evaluate dispatch or scheduling quality rather than whether their learned graph or attention weights can be presented as faithful explanations when vehicle telemetry becomes stale. They provide useful task benchmarks, but not direct faithfulness benchmarks.

2.2 Explainability in reinforcement learning

Explainable reinforcement learning (XRL) covers more than a visual saliency map. An explanation may describe which state features mattered, summarize a policy, contrast the selected action with an alternative, or show the sequence of events that led to an outcome. Reviews by Heuillet, Couthouis and Díaz-Rodríguez (2021) and Milani et al. (2024) stress that the method must match the user's question. A model developer debugging a policy and an operator checking one live action do not need the same output. The systematic taxonomy by Bekkemoen (2024) reaches the same conclusion across a larger body of recent XRL studies.

This dissertation concerns a local, post-decision operator question: *which visible vehicles or requests supported the selected action, whether dispatch or no-op?* Earlier XRL work offers several ways to answer this question. Greydanus et al. (2018) perturb image regions to visualize what changes an Atari policy. Madumal et al. (2020) use a causal model to generate contrastive explanations. Mott et al. (2019) introduce an attention bottleneck intended to expose what an RL agent uses. These approaches differ in mechanism, but they share an important lesson: a useful picture does not necessarily show what the model relied on.

For an operational dashboard, readability and faithfulness must be considered separately. An explanation may be easy to read but weakly connected to the policy computation. Perturbation tests assess whether removing the identified evidence weakens the prediction and whether retaining sufficient evidence preserves it (DeYoung et al., 2020). This experiment tests that link through a controlled decision-level faithfulness test; it does not measure user preference for the attention map.

Recent XRL work also clarifies which question an explanation answers. Causal state distillation (W. Lu et al., 2024) extends reward decomposition with objectives for sparse, sufficient and distinct causal factors. This is a model-design route to explanatory structure, whereas the present study audits a frozen policy with no explanatory training objective. COViz (Amitai, Septon and Amir, 2024) displays the outcomes of chosen and alternative actions and evaluates human understanding. It addresses the consequences of a choice rather than the importance of each input node. These approaches are complementary: perturbation faithfulness cannot establish that a dispatcher understands an explanation, and a useful counterfactual visualization cannot by itself validate raw attention.

2.3 Is attention an explanation?

A widely cited challenge to attention-based explanations was presented by Jain and Wallace (2019). They found that attention weights could be weakly correlated with other importance measures and that very different attention distributions could produce similar predictions. Wiegreffe and Pinter (2019) argued that this does not justify rejecting all attention explanations and proposed more careful tests. Serrano and Smith (2019) likewise showed that removing high-attention items can affect outputs, but that the relationship varies by model and task. The literature therefore supports neither a universal acceptance nor a universal rejection of attention. Each explanation claim needs an explicit definition and an appropriate test.

Attention became widely visible through Transformer models (Vaswani et al., 2017), but visibility also encouraged a broader interpretability claim than the mechanism alone can support. Lipton (2018) warns that the word *interpretability* often combines several different goals that should be evaluated separately.

Jacovi and Goldberg (2020) separate *plausibility*, which concerns whether an explanation looks reasonable to a person, from *faithfulness*, which concerns whether it reflects the model behavior that produced the output. This dissertation follows that distinction. It treats the attention weights as a candidate explanation and measures their decision relevance rather than accepting them simply because they are part of the actor.

Layer and head aggregation add another difficulty. A two-layer, multi-head network does not produce one unique attention map. Averaging heads, selecting a layer, taking a maximum, or composing attention across layers can produce different rankings. Abnar and Zuidema (2020) propose attention rollout and flow to account for information mixing across layers, and report stronger correlations with ablation and gradient importance than raw attention in Transformers. Although their model is not a GAT-MARL dispatcher, the methodological point transfers: an attention result is incomplete unless the aggregation rule is defined and its sensitivity is checked. Shin et al. (2025) similarly show that raw self-attention coefficients can give weak graph attributions when message-passing paths are ignored.

2.4 Explaining graph neural networks

GNN explanations may identify important features, nodes, edges, or subgraphs. GNNExplainer learns a compact subgraph and feature mask that preserves a prediction (Ying et al., 2019). PGExplainer parameterizes explanation generation so that it can be shared across examples (Luo et al., 2020). SubgraphX searches for important subgraphs using Shapley-value approximations (Yuan et al., 2021). These methods show that graph explanation is a distinct problem: removing one node can alter both its own features and the messages available to other nodes.

The survey by Yuan et al. (2023) organizes GNN explainers by target, mechanism, and scope. GraphFramEx goes further by comparing explanation methods under different user needs and combining necessity and sufficiency views (Amara et al., 2022). Its results show that no single method dominates every evaluation dimension. This supports the use of several checks in the present study: DEF tests counterfactual relevance, taxi-only rank correlation compares attention with leave-one-out effects, and stale-attention measures describe freshness exposure. Azzolin et al. (2025) further show that GNN faithfulness metrics are not interchangeable, reinforcing the need to define what each measure tests.

Most GNN explainability benchmarks study node or graph classification. Fleet dispatch differs because some graph nodes also define actions. A passenger request is both information and a selectable action, while a nearby taxi is context only. Deleting them creates different interventions. This action-linked graph structure motivates the construct-validity audit in Section 3.7.

Several 2024-2025 methods offer alternatives to raw attention. GOAt (S. Lu et al., 2024) analytically allocates graph outputs to input features and evaluates fidelity, discriminability and stability. Its explicit attribution path is attractive, but transferring it to a residual attention actor requires accounting for the actual computation and selected output. Bui et al. (2024) use the Myerson-Taylor interaction index to respect graph connectivity and interactions. This exposes a limitation of a single-node ranking: two jointly relevant nodes need not have large individual LOO effects. Applying either approach to dispatch would require accounting for request nodes that also define available actions.

The output type matters as well. RegExplainer (Zhang et al., 2024) adapts graph explanations to regression through information-bottleneck, mix-up and self-supervised objectives. It challenges the assumption that a classification-oriented explanation score transfers unchanged to a continuous value estimate. Here, the target is the actor's selected action, not the critic's team value; that distinction determines which output and perturbation to evaluate.

Global explanations answer a different question from the present local audit. GraphTrail (Armgaan et al., 2024) converts predictions into logical rules over mined graph concepts, while GNNBoundary (Wang and Shen, 2024) generates graphs near class boundaries. Both reveal model-level behavior that a local ranking can miss. However, global agreement or a plausible boundary graph cannot establish that one particular dispatch decision used fresh vehicle data. Yu and Gao's MAGE (Yu and Gao, 2025) generates motif-based molecular explanations, emphasizing domain-valid substructures. Its molecular validity requirements illustrate why a dispatch explainer likewise needs task-specific constraints. This MAGE is distinct from the Myerson-Taylor explainer of Bui et al. (2024).

GraphNarrator (Pan et al., 2025) generates textual explanations using saliency-derived pseudo-labels and iterative refinement. This makes explanations easier to present, but their quality still depends on the underlying attribution: a fluent explanation cannot serve as an independent check of the saliency labels that helped train it. For this dissertation, adding text to an attention map would therefore require the same freshness and decision-relevance audit.

2.5 Evaluating faithfulness: methods and pitfalls

Comprehensiveness and sufficiency are common perturbation measures. In the ERASER benchmark, comprehensiveness asks how much a prediction weakens when the explanation is removed, while sufficiency asks how well the selected evidence alone preserves it (DeYoung et al., 2020). Liu et al. (2022) similarly test attention by measuring faithfulness violations. These measures are useful because they connect an explanation to model behavior instead of visual appeal.

Model-agnostic methods such as LIME also use local perturbations to estimate which inputs support a prediction (Ribeiro, Singh and Guestrin, 2016). Their usefulness depends on whether the perturbed samples remain meaningful for the model and task.

Perturbation is not automatically valid. Removing input features may create samples unlike the data seen during training. The Remove and Retrain (ROAR) benchmark shows why deletion can confound attribution quality with distribution shift (Hooker et al., 2019). Zheng et al. (2024) identify this problem in GNN fidelity measures and propose more robust alternatives. Adebayo et al. (2018) test whether saliency methods depend on learned parameters and training data by randomizing model parameters and retraining on randomized labels. Li et al. (2024) show that small graph changes can preserve a prediction while substantially changing its explanation. Together, these studies show that the evaluator itself must be audited.

The present action-deletion problem is related but more specific. If a selected request node is masked, its action logit becomes unavailable. A large margin change can then occur even if the request features carried no useful evidence. A random baseline that removes more request nodes than the attention top-k is not a fair comparison. Type matching, selected-action protection, and clamp logging are introduced to control this mechanism. A leave-one-out perturbation ranking acts as a positive control for comprehensiveness, while Gradient x Input (Ancona et al., 2018) provides a comparator that does not use attention coefficients.

The small graph creates a second limit. When k = 3 and only a few valid nodes exist, a random size-three set must overlap substantially with the explanation set. This compresses the possible difference between them. Reporting the exact expected overlap and a taxi-only rank correlation makes that limitation visible instead of treating every near-zero DEF as decisive evidence.

Recent evaluation research provides stronger alternatives to a single fidelity number. He et al. (2025) establish connections between edge-gradient and perturbation methods, including equivalence under restricted linear settings. This suggests that agreement between two explainers may reflect shared mechanics rather than independent validation; their conditions should not be assumed for the nonlinear actor used here. PowerGraph (Varbella et al., 2024) supplies cascading-failure explanation ground truth in a domain-specific benchmark. It demonstrates the value of external explanatory labels, which the present SUMO experiment lacks: its LOO ranking remains a positive control, not ground truth. Saha and Bandyopadhyay (2026) introduce sufficiency-risk certificates and coverage, gain and overlap measures for model-level graph explanations. Their uncertainty-aware evaluation is relevant to audit design, but those certificates do not automatically apply to a local action-level DEF score.

2.6 Age of Information and telemetry degradation

Age of Information (AoI) measures the time since the newest received update was generated (Kaul, Yates and Gruteser, 2012). It is widely used to study communication freshness, including the relationship between update age, estimation, and control (Yates et al., 2021). The explanation question is different. A dispatch action and its attention map may remain available even when a vehicle reading has stopped updating. The operator can see a strong weight without knowing that its source is old.

Delayed-observation reinforcement learning studies how an agent can continue to act when observations or actions arrive late. Bouteiller et al. (2021) model random action and observation delays and propose a delay-correcting actor-critic method. Liotet et al. (2022) learn a delayed policy from demonstrations produced without delay. Both studies focus on maintaining decision performance under delay. They do not test whether an explanation remains faithful when the policy receives stale telemetry.

Recent delay-aware RL explicitly models uncertainty about the current state. Yao, Florescu and Lee (2024) use stochastic planning for delayed feedback, whereas Wu et al. (2025) directly forecast belief to reduce errors accumulated by recursive state prediction. Both provide candidate approaches to recovering useful state information; neither result establishes that an attention ranking is faithful to the resulting action. Zhou et al. (2024) instead make observation timing part of the action. This separates a deliberately scheduled observation from an externally triggered outage. The present freeze intervention models the latter and does not optimize the communication schedule.

Communication robustness also differs from explanation robustness. MAGI (Ding et al., 2024) uses a graph information bottleneck to learn compact, action-relevant messages under perturbations. Soudijani and Dimitrova (2025) jointly synthesize action and communication policies subject to communication restrictions. These methods address information exchange and task objectives; robust coordination is not itself an explanation-release criterion. In vehicle perception, MRCNet (Hong et al., 2024) addresses motion and sensing noise during communication and fusion. Its setting motivates testing richer telemetry failures, although perception noise and a last-value freeze are distinct interventions.

Dynamic graph attribution is especially close to the present problem. Liu and Xie (2025) attribute changes in predictions on evolving graphs to message flows and layer edges, with KL-divergence used to select explanatory edges. This supports explicitly comparing two observations rather than treating a static map as sufficient. Their evolving edge weights differ from the feature-level freeze with fixed paired node identities here. Liu et al. (2025) use graph curvature and resistance to improve explanation robustness without retraining the target model. That is a potential improvement to an explainer, whereas this study measures whether the declared raw-attention channel is stable. Robustness to structural perturbations does not establish robustness to stale features, so the two require separate tests.

This dissertation uses AoI only to describe the freshness of each node and as the weight inside weighted attention mass on stale nodes (WAMSN). The experiment manipulates the duration of an observation-layer outage triggered by tunnel entry, not AoI itself. It therefore makes no causal claim that a larger AoI value lowers faithfulness. The analysis instead asks whether stale exposure and attention-based explanation behavior provide consistent evidence under controlled outages.

2.7 Deciding when an explanation can be shown

Explanation tests can inform the decision to show an explanation to a user. The NIST AI Risk Management Framework treats testing, validation, documentation, and monitoring as parts of managing the risks of AI systems (Tabassi, 2023). It provides general risk management guidance rather than a test for attention explanations.

This dissertation applies that principle to a specific question: whether an attention map has enough supporting evidence to be presented for a fleet-dispatch decision. The resulting framework checks the data status, the validity of the faithfulness test, and consistency across independently trained policies. It then returns one of three decisions: *ELIGIBLE*, *WITHHOLD*, or *INCOMPLETE*. These labels and their decision rules are proposed in this dissertation; they are not prescribed by the NIST framework. Recent work also shows why such caution is needed: Azzolin et al. (2026) show that a self-explaining GNN can produce an explanation unrelated to its prediction process and that some metrics may miss this failure. Although their setting is graph classification, the result supports auditing an explanation before presenting it as evidence.

2.8 Research gap

The literature supports four points: relational RL is appropriate for fleet dispatch; attention is easy to expose but disputed as an explanation; GNN faithfulness requires graph-aware perturbations; and stale state matters to operational control. Recent work examines GNN explanation fidelity, instability, and auditing (Zheng et al., 2024; Li et al., 2024; Azzolin et al., 2025; Shin et al., 2025; Azzolin et al., 2026), but does not combine MARL fleet dispatch, paired clean and stale observations, and action-linked request nodes. The present study brings these elements together in a controlled evaluation.

The closest task studies are BMG-Q (Hu, Feng and Li, 2025), CoopRide (Wang et al., 2025) and DualG-MARL (Sha et al., 2026). Their dispatch objectives and datasets differ from this controlled benchmark, so their performance figures are not directly comparable. The closest methodological studies instead concern attribution paths, fidelity measurement, explanation instability and changes in graph inputs. Table 2.1 compares eight closely related studies, highlighting their contributions and the adaptations needed for the dispatch setting.

Across the graph- and RL-explainability and delayed-observation studies reviewed in Sections 2.2-2.7, no evaluation protocol was identified that combines all of the following:

- a graph-attention MARL action explained at decision level;
- paired clean and stale versions of the same operational observation;
- explicit measurement of attention placed on stale telemetry;
- a control for graph nodes that also remove available actions;
- replication across independently trained checkpoints.

The proposed audit protocol uses the controlled state available in SUMO (Lopez et al., 2018) and the attention weights exposed by GAT-MAPPO to test explanation faithfulness and sensitivity to stale telemetry. It links these measurements to a decision about whether the evidence is sufficient to present the attention map to an operator.

Table 2.1: Critical comparison of eight closely related explanation studies

Study	Method and setting	Evaluation and finding	Strength	Limit and implication for this study
Zheng et al. (2024)	Robust fidelity for subgraph explanations	Synthetic and real graphs; conventional deletion can distort fidelity	Explicit treatment of distribution shift	Does not resolve request-linked action removal; motivates separate action protection
Li et al. (2024)	Adversarial changes to graph structure	Explanations can change while predictions remain correct	Separates prediction and explanation robustness	Adversarial edges differ from stale vehicle features; requires a paired telemetry test
Azzolin et al. (2025)	Analysis of regular and self-explainable GNNs	Faithfulness metrics need not agree	Exposes dependence on metric and architecture	Prevents interpreting DEF as a universal or complete causal score
Shin et al. (2025)	GAtt computation-tree edge attribution	Synthetic and real data; improves on naive attention attribution	Accounts for message-passing paths	Edge attribution needs adaptation to residual actor outputs and action-linked nodes
Liu and Xie (2025)	Layer-edge attribution on evolving graphs	Eight datasets; evaluates fidelity of prediction changes	Directly studies change between graphs	Evolving edge weights differ from the fixed-identity telemetry twins used here
Liu et al. (2025)	Curvature and resistance regularization	Nine datasets across three graph tasks; improves robustness	Leaves the target model unchanged	Structural robustness does not establish stale-feature or action-level faithfulness
Azzolin et al. (2026)	Audit of degenerate self-explanations	Accurate models can yield explanations unrelated to prediction	Shows why accuracy is insufficient for assurance	Self-explainable classifiers differ from raw-attention MARL; motivates an explicit rejection path
Saha and Bandyopadhyay (2026)	Certified model-level evaluation	Sufficiency risk, coverage, gain and overlap with uncertainty	Evaluates quality beyond target-class score	Model-level certificates are not local DEF guarantees; motivates scoped statistical claims

Chapter 3: Methodology

3.1 Study design and proposed approach

The study uses a controlled simulation because the research question requires the same decision to be observed in two ways: once with current telemetry and once with degraded telemetry. A conventional observational fleet log would not normally provide this exact clean twin. SUMO provides the simulated physical state, while a separate observation layer controls what the policy receives.

The protocol is defined in version-controlled configuration files. It contains three stages: training, validation-based checkpoint selection, and held-out evaluation. Faithfulness sweeps are run only after the selected checkpoints are frozen.

The proposed approach has two connected paths. SUMO ground truth passes through an observation-layer degradation mechanism into the local graph and GAT-MAPPO actor. A separate, read-only path extracts attention, constructs paired clean twins, evaluates action-aware faithfulness and aggregates the evidence into an explanation-release decision. Figure 3.4 shows the actor and attention channel; Figure 5.2 shows the offline release workflow. The methodological contribution lies in the paired evaluation and release protocol, with GAT-MAPPO serving as the policy under audit.

Each stage addresses a specific failure mode. The clean twin controls the underlying world state; per-type embeddings distinguish vehicles from requests; the graph encoder combines relational context; the actor identifies the action to explain; and the centralized critic supports learning without being needed for decentralized action selection. AoI records freshness without assuming that a large attention weight means a fresh reading. Type matching and selected-action protection remove a known perturbation confound. The final audit combines evidence that no single score supplies. Table 3.10 summarizes the implementation interfaces and their purpose.

3.2 Dataset selection and exploratory data analysis

The road network is an OpenStreetMap extract of Central Park in Yubei, Chongqing. Map data are credited to OpenStreetMap contributors under the Open Database License (OpenStreetMap contributors, n.d.). The committed XML records extraction and network conversion on 22 June 2026 using SUMO tools 1.27.0. This network-building version is distinct from the recorded simulation runtime in Section 3.11.1. Roads marked as tunnels supply physically meaningful signal-loss triggers. SUMO simulates 20 taxis and 50 passenger requests during each 1,200-second episode. Taxi dispatch is controlled through TraCI, while SUMO continues to update the true position and speed of every vehicle.

Twenty generated demand variants use one fixed taxi seed (42) and passenger request seeds 1000-1019. Variants 0-13 form the training split, variants 14-16 form the validation split, and variants 17-19 form the held-out test split. The policy is fitted on the 14 training variants. Candidate checkpoints are compared on the three validation variants. The final reported runs use only the three test variants. This separation prevents test results from influencing model selection. During evaluation, consecutive episodes cycle through the three test variants, and actions are sampled from the policy's probability distribution. Evaluation seeds control this action sampling; they do not create additional independent demand datasets. The same demand splits were used in an earlier pilot study. They remain separate from training and checkpoint selection, but this rerun does not constitute validation on a newly collected dataset.

SUMO generates microscopic traffic simulations (Lopez et al., 2018) from the chosen road network, demand, routes, vehicle behavior, tunnel locations, and random seeds. These modeling choices shape the results and are held constant in the paired comparison. At each sampled decision, the clean and degraded observations share the same underlying SUMO state and trained-policy checkpoint; only the observation-layer telemetry differs. This controls the comparison well enough to isolate the effect of the implemented degradation within this scenario, but it does not make the simulated fleet representative of every real city.

3.2.1 Data generation and selection rationale

SUMO was selected because the study needs an interactive traffic environment with access to the physical state of each vehicle (Lopez et al., 2018). Through TraCI, the implementation reads vehicle states and applies taxi-dispatch actions at each decision step. A separate observation layer can then retain outdated readings while vehicles continue moving in the simulation. This separation allows the same decision state to be evaluated with current and degraded telemetry, which is central to the paired explanation audit. Fixed road, demand and seed inputs also support repeatable evaluations across policies and outage settings.

The committed demand generator samples distinct origin and destination edge identifiers for each request and an integer arrival time uniformly from 0 to 600 seconds. Requests therefore arrive during the first half of the episode; the remaining time permits ongoing journeys to finish. Edge sampling is not weighted by population, road length or observed passenger demand. The 20 variants contain 1,000 generated requests in total: 700 training, 150 validation and 150 test requests. These are scenario inputs, not 1,000 independently observed real-world journeys. The fixed taxi placement and shared road network make the split a demand-generalization test within one scenario.

The Central Park network is suitable for a controlled outage study because its tunnel metadata supplies localized, repeatable trigger edges while the surrounding roads permit clean and exposed decisions in the same network. The configured trigger set contains three edges; the metadata also excludes orphan tunnel edges that would not support meaningful traversals. The scenario was chosen for these features; it is not a representative sample of Chongqing. Twenty taxis, 50 requests and a 1,200-second horizon define a small, manageable benchmark for repeated paired counterfactual evaluation. These settings are experimental design choices, not estimates of real fleet size or demand. The demand-generation assumptions, random-circling idle taxis and configured vehicle dimensions must therefore be considered when transferring the results.

3.2.2 Spatial and temporal exploration

The EDA describes the fixed demand splits using the committed route XML files and road geometry. It was conducted after evaluation and was not used for model selection or hyperparameter tuning. Figure 3.1 plots pickup and drop-off edge midpoints against the road network and the configured tunnel triggers. Locations are proxies obtained from the midpoint along the first lane's polyline, not observed passenger coordinates or an exact SUMO boarding point.

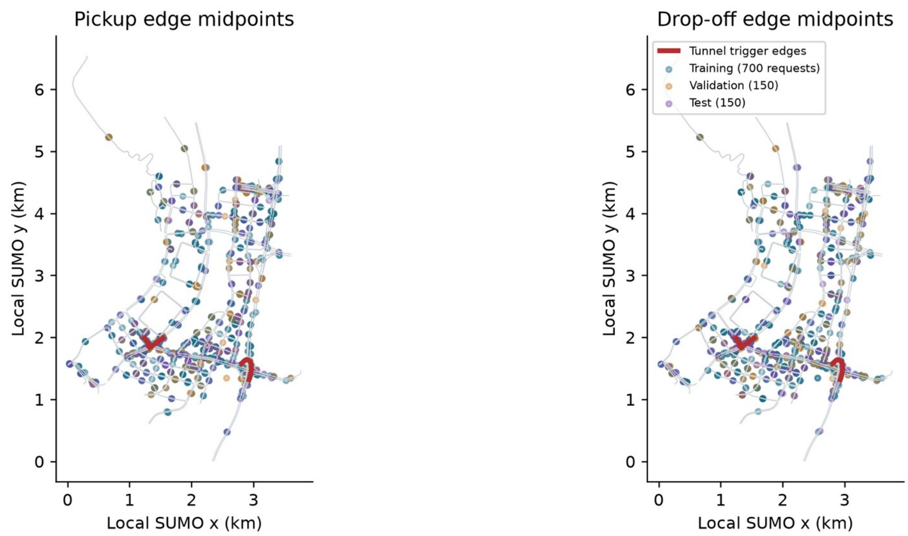

Figure 3.1. Spatial EDA of all 20 demand variants. Grey lines are non-internal road edges, red lines are the three configured tunnel-trigger edges, and colored points distinguish training, validation and test demand. Map data: OpenStreetMap contributors, ODbL; https://www.openstreetmap.org/copyright. Coordinates use the local SUMO frame. Repeated edge choices can overlap on the map.

Mean request-arrival times are 289.8, 300.8 and 294.1 seconds for training, validation and test, respectively. Their ranges are 0-600, 5-599 and 0-598 seconds. Figure 3.2 compares empirical cumulative distributions so that the unequal split sizes do not distort the comparison. Arrivals occupy broadly similar portions of the generation window, but this does not establish statistical equivalence between the small validation and test samples.

The median straight-line origin-destination separation is 1.558 km in training, 1.495 km in validation and 1.302 km in test. The corresponding 10th-90th percentile intervals are 0.502-2.909, 0.537-2.967 and 0.364-3.055 km. Test demand therefore has a lower median separation despite a comparable upper tail. This may affect journey completion and reinforces using identical held-out demand for policy comparisons. The distance is a Euclidean proxy between edge midpoints; it is not a routed distance, congestion estimate or observed travel time.

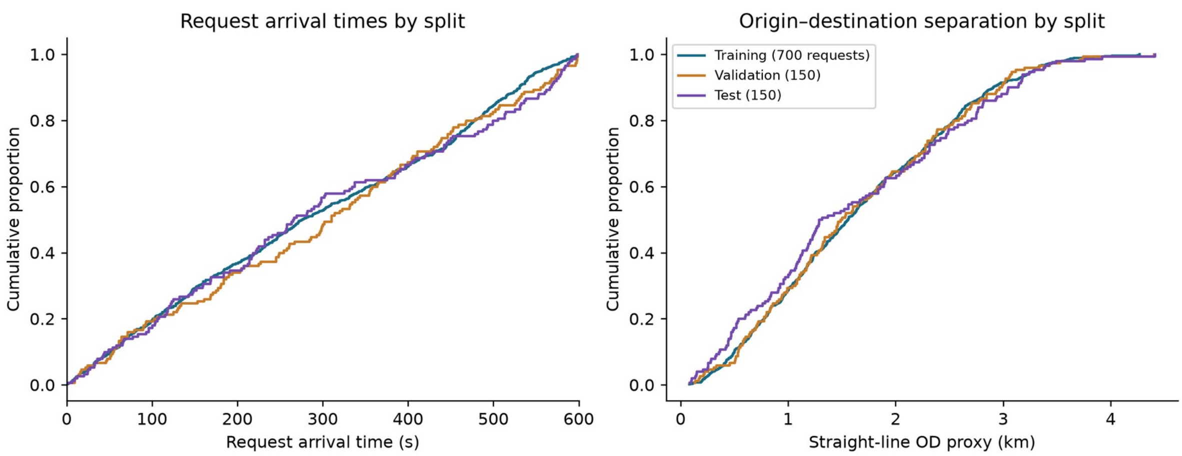

Figure 3.2. Demand-split EDA. Left: request-arrival times. Right: straight-line origin-destination separation. Requests are generated during the first 600 seconds of each 1,200-second episode; the remaining time allows ongoing journeys to finish. Each curve is normalized within its split; sample sizes are 700, 150 and 150 requests.

The retained route files each contain 20 taxi trips and 50 requests, and every request origin and destination resolves to an edge in the committed network. The generator and manifest make the split reproducible, but sampling different requests on one map does not test unseen road layouts or real demand shifts. No rebalancing or split changes were made in response to this descriptive EDA.

3.2.3 Policy-conditioned observations and exposure

The graph and telemetry records depend on the policy trajectory, unlike the fixed input demand above. Table 4.4 reports their valid-node distribution: the scored observations contain about five peer taxis and 2.5 request nodes on average. Padding must therefore be masked, and a fixed top-three explanation may cover a substantial fraction of the eligible graph. Figures 4.7 and 4.9 report outage-conditioned WAMSN and action composition, respectively. These describe the observations actually scored, not the unconditional distribution of all simulation steps. Exposure-triggered sampling scores stale observations more densely than clean observations, so record proportions must not be interpreted as fleet-wide outage prevalence. The paired analysis and episode-level resampling account for the declared comparison, but they do not make the scored records independent data samples.

3.3 Observation graph and action space

Each idle taxi receives a self-centered graph. Table 3.1 summarizes its node types, features, and roles.

Table 3.1: Observation graph node types and roles

Node type	Main features	Role
Self taxi	position, normalized time, speed and AoI; separate position-valid indicator	acting vehicle
Peer taxi	relative position, availability, distance, AoI	nearby fleet context
Passenger request	pickup/drop-off vectors, waiting time	candidate request and action

The graph contains up to five nearby taxis and five nearby requests. Features are normalized, and request and taxi masks prevent padded nodes from receiving attention. The discrete action space contains no-op plus one action for each visible passenger request. Figure 3.3 shows the maximum graph size and the request-to-action mapping. This mapping is important for the faithfulness evaluation: masking a passenger request node also removes its associated action, whereas masking a peer-taxi node removes only information about that taxi. Section 3.7 explains how the evaluation accounts for the different effects of masking request and peer-taxi nodes.

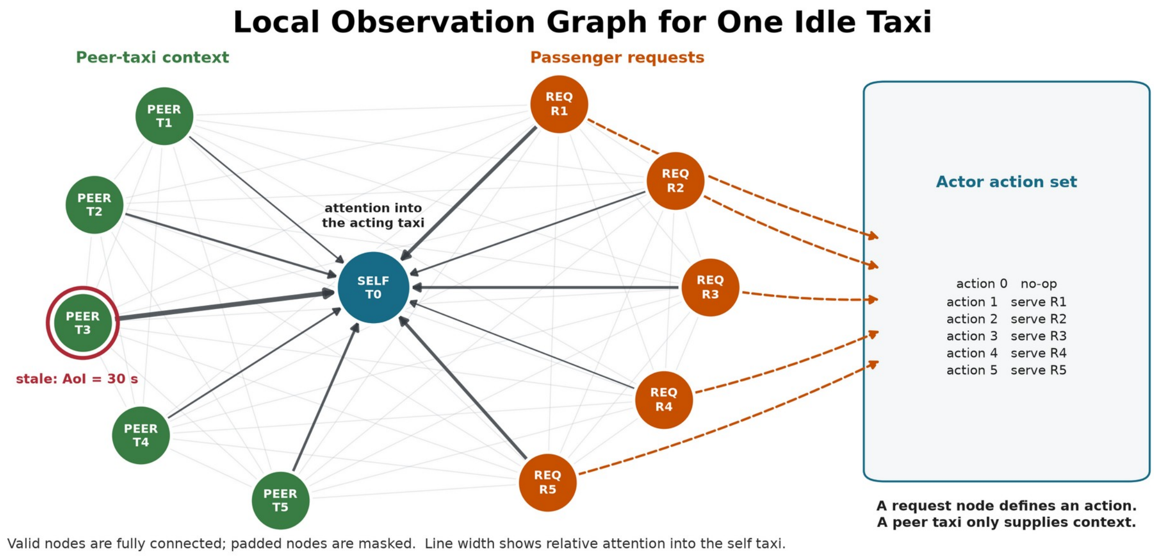

Figure 3.3. Maximum configured graph: one self taxi, five peer taxis, and five requests. Valid nodes are fully connected, and unused slots are masked. The red ring marks stale telemetry; dark edges show illustrative attention into the self taxi. Each request maps to one dispatch action.

3.4 Policy models and training

The main policy uses per-node-type feature projections followed by two graph attention layers with four heads. The actor scores no-op and the visible requests. The critic estimates value for MAPPO training (Yu et al., 2022) with centralized training and decentralized execution (CTDE). The GAT implementation returns its attention tensors directly so they can be audited without changing the trained network.

During training, the centralized critic pools the valid node embeddings from all agents in the same transition and returns one shared team-value estimate. The actor does not receive this pooled representation: each taxi selects its action from its own local graph. At execution and during the explanation audit, the actor therefore remains decentralized.

The implementation uses scaled dot-product graph attention on the fully connected set of valid local nodes. This differs from the additive attention score in the original GAT paper, so "GAT" in the experiment labels denotes the graph-attention policy family, not an exact reproduction of that layer. For node i, node j, layer l, and head h:

$$
\begin{gathered}
q_i = W_{Q,h}\,\mathrm{LayerNorm}(h_i^l)\\
k_j = W_{K,h}\,\mathrm{LayerNorm}(h_j^l)\\
v_j = W_{V,h}\,\mathrm{LayerNorm}(h_j^l)\\
s_{ij}^{l,h} = \frac{q_i k_j}{\sqrt{d_{\mathrm{head}}}}\\
\alpha_{ij}^{l,h} = \mathrm{softmax}_j(s_{ij}^{l,h})\\
z_i = \mathrm{concat}_h(\Sigma_j(\alpha_{ij}^{l,h} \times v_j))\\
u_i = h_i^l + W_O z_i\\
h_i^{l+1} = u_i + \mathrm{FFN}(\mathrm{LayerNorm}(u_i))
\end{gathered}
\tag{3.1}
$$

Here, FFN denotes the feed-forward network within each attention layer. The mask removes padded nodes before the softmax. A residual connection keeps the previous node embedding.

The actor reads the updated self embedding for no-op and each updated request embedding for its matching dispatch action. The default explanation is the self row (i = 0) averaged over both layers and all four heads, then renormalized:

$$\alpha_j = \text{normalize}\left(\frac{\Sigma_{l,h}\left(\alpha^{l,h}_{0j}\right)}{LH}\right) \tag{3.2}$$

This aggregation is declared before evaluation. A sensitivity analysis also reports each layer, head-wise values, head maximums, and attention rollout; it does not select an aggregation after seeing which result is favorable. The explanation average is an analysis choice: inside the policy, head outputs are concatenated and projected rather than averaged.

The fixed self row represents the acting taxi's dashboard view and is the explanation channel originally declared for audit. It directly contributes to the no-op embedding. A request-action logit, however, is read from that request's updated embedding, so its corresponding request row is more directly aligned with the selected logit. A sensitivity analysis therefore reports a separate request-action sensitivity comparison between the fixed self row and the selected request row. This comparison is not used to redefine the primary metric after seeing the result.

Figure 3.4 summarizes the policy path and the read-only audit channel.

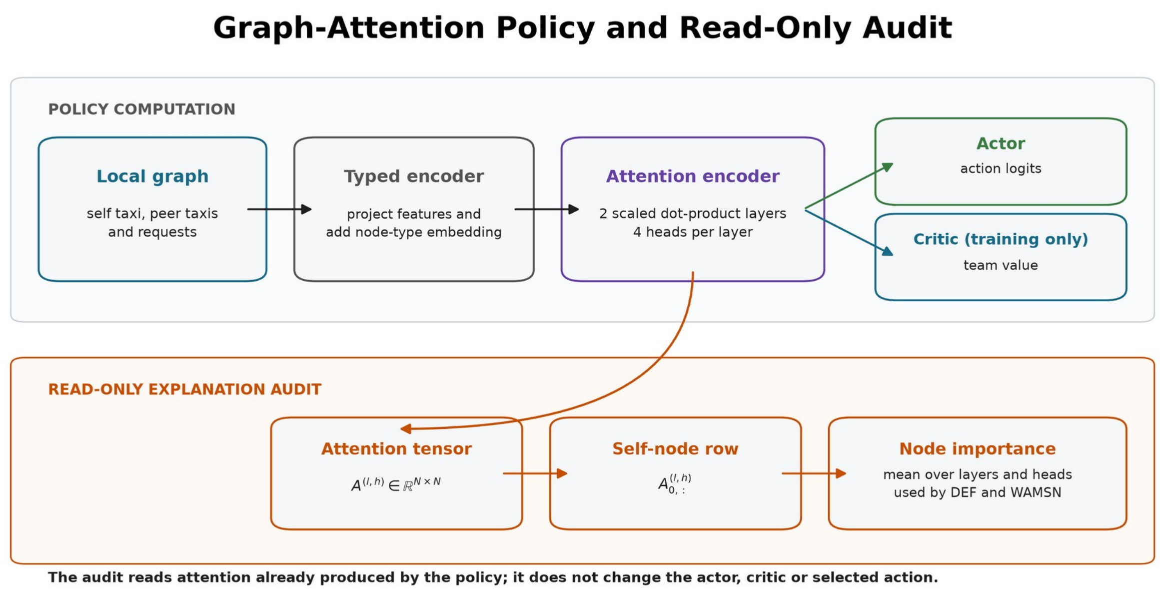

Figure 3.4. The policy and audit share the same two-layer attention encoder. The default audit reduces the self-node attention row to node importance without modifying the actor, critic, or selected action; the selected-request row is evaluated separately as a sensitivity check.

Table 3.2 lists the three policy conditions included in the experiment. The non-graph baseline uses a multilayer perceptron (MLP).

Table 3.2: Policy conditions used in the experiment

Model	Policy	Training observations	Purpose
MLP	MLP-MAPPO	clean	performance context without graph attention
GAT	GAT-MAPPO	clean	primary attention model under audit
GAT-Outage	GAT-MAPPO	tunnel-triggered 30-second outages	tested degradation-aware training configuration

Each model is trained in five independently initialized runs using seeds 42-46. All three models receive 50 training epochs. GAT and GAT-Outage use the same learning rate, linear decay schedule, PPO settings and checkpoint-selection criterion; MLP retains its architecture-specific learning rate. Corresponding GAT and GAT-Outage seeds are paired in the H5 comparison.

Equal epoch budgets do not guarantee equal numbers of agent decisions or PPO updates because the learned policies follow different trajectories. Training logs retain those counts. Validation telemetry matches each model's training condition, as specified below. H5 therefore compares the fitted clean and outage training-and-selection configurations, rather than isolating training telemetry as the only difference.

Candidate checkpoints include periodic trained snapshots, the final snapshot, and a rolling-best snapshot that may come from any trained epoch. The epoch-0 snapshot is saved after the first training epoch. It is evaluated only as an early-training diagnostic and is excluded from final checkpoint selection. Because the implementation records zero-based indices, index 49 denotes the checkpoint saved after 50 training epochs. Selection uses eight sampled-action validation episodes with seed 2026 and the mean number of completed passenger journeys as the primary criterion; mean reward and the earlier checkpoint index break ties. MLP and GAT are selected under clean validation observations. GAT-Outage is selected under its 30-second tunnel-triggered training condition. The test split is not read during selection. Table 3.3 reports the selected checkpoint index for each training run.

Table 3.3: Validation-selected checkpoint indices (zero-based)

Model	seed 42	seed 43	seed 44	seed 45	seed 46
MLP	20	20	40	40	49
GAT	40	28	10	20	10
GAT-Outage	10	10	30	20	31

Table 3.4 reports the training and optimization settings.

Table 3.4: Training and optimization settings

Setting	Value
Training epochs	50 for all models
Learning rate	0.0001 (GAT variants), 0.0003 (MLP)
Learning-rate schedule	linear decay over each run (GAT variants); constant (MLP)
Discount factor (gamma)	0.99
Generalized advantage estimation factor (lambda)	0.95
PPO clip ratio	0.20
PPO passes per update	4
Minibatch size	256
Value-loss coefficient	0.25
Value clip ratio	0.20
Entropy coefficient	0.05, adaptive to a 0.20 entropy floor
Maximum adaptive entropy coefficient	0.20
Target Kullback-Leibler divergence	0.015
Gradient-norm limit	0.50

Setting	Value
Dispatch credit reward	1.00
Critic encoder gradient scale	0.10
GAT hidden width / layers / heads	64 / 2 / 4
GAT attention dropout	0
MLP hidden width / layers	176 / 3

Table 3.5 separates checkpoint acceptance from the optimizer settings. Final validation retention is the final checkpoint's mean validation completed journeys divided by the selected checkpoint's mean validation completed journeys. The stability window defines where the training measures are calculated; the remaining four gates must pass before a training run is accepted for evaluation.

Table 3.5: Checkpoint-acceptance criteria

Checkpoint-acceptance setting	Required value
Stability window	Final 10 training epochs
Minimum final-window mean completed journeys	5.0
Maximum consecutive zero-completion epochs	4
Minimum selected validation completed journeys	5.0
Minimum final validation retention	0.80

At simulation step t, the reported shared team reward is:

$$r_t = 10\, N_{\text{complete},t} + 0.5\, N_{\text{dispatch},t} - 0.001\, \text{mean\_wait}_t \tag{3.3}$$

For PPO training, taxi i receives an additional difference credit only when its dispatch is accepted:

$$r_{i,t}^{\text{train}} = r_t + 1.0\, I_{\text{successful dispatch by i at t}} \tag{3.4}$$

Here, N_complete,t counts target passengers who reach their destinations; it is not a boarding count. I(.) is one when the condition is true and zero otherwise. A taxi can be absent from the action set while it serves a request. The implementation therefore keeps its last decision open and accumulates the shared reward from every intervening simulation step. For a gap of Delta steps, the interval return discounts those rewards by gamma^j, bootstraps with gamma^Delta, and applies the GAE trace factor (gamma lambda)^Delta. Terminal intervals do not bootstrap into the next episode. This semi-Markov treatment prevents rewards earned while a taxi is busy from disappearing. Neither reward contains a term for a faithful, stable, or human-readable attention map.

Figure 3.5 shows how training, validation selection, and held-out evaluation remain separated.

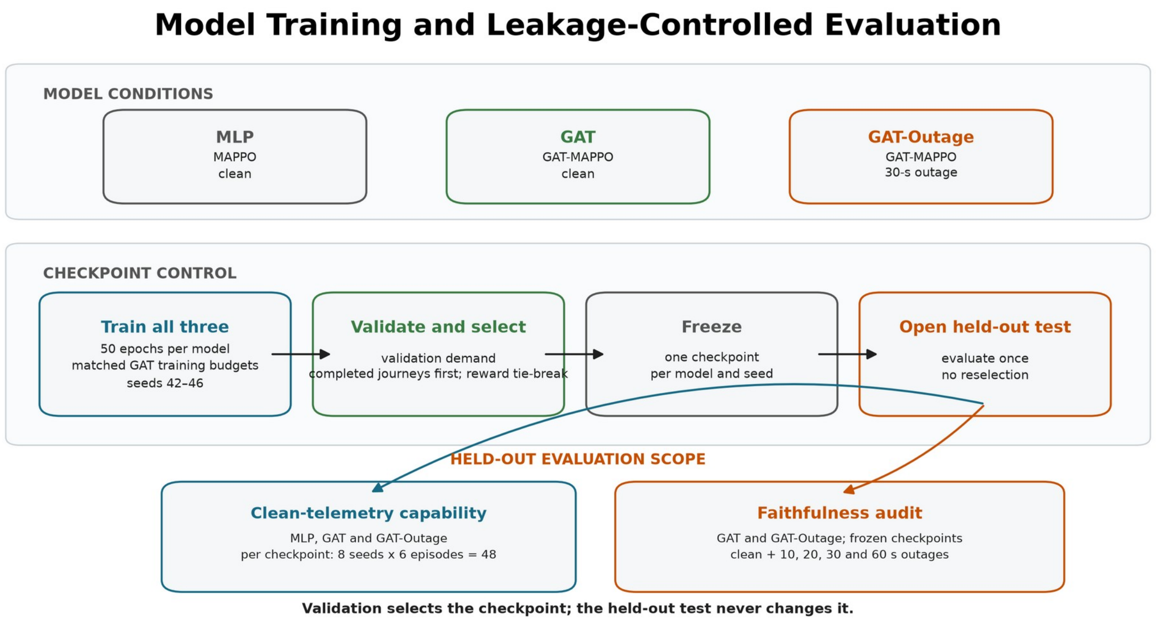

Figure 3.5. Shared training and validation-based selection pipeline with the common 50-epoch budget and model-specific settings described above. Each frozen checkpoint receives 48 clean held-out episodes. The faithfulness audit then compares the frozen GAT and GAT-Outage checkpoints across clean and four outage conditions without reselection. This produces 1,200 faithfulness-sweep episodes per GAT model and 2,400 in total.

3.5 Telemetry degradation

The trigger and the degradation location are separate parts of the design.

Step 1 - Trigger: when a taxi enters a tunnel edge, the event starts one outage.

Step 2 - Observation-layer outage: for the declared duration, the policy sees the taxi's last valid position and speed. SUMO continues to simulate the true state.

Step 3 - Recovery: after the fixed window expires, a current reading is accepted. A taxi must leave and enter a tunnel again before another tunnel-entry event can trigger.

The fixed outage durations are 10, 20, 30, and 60 seconds. They are easy to monitor and compare, and they create increasing empirical degradation rates. A longer outage is not assumed to cause a larger loss of faithfulness.

AoI is current simulation time minus the last valid update's timestamp (Yates et al., 2021) and is included in each GAT observation. WAMSN uses normalized AoI as a staleness weight. For this reason, an increase in WAMSN with outage duration partly verifies that the manipulation created longer stale exposure; it is not by itself proof that AoI changed DEF. Figure 3.6 illustrates this separation between physical state and policy observation.

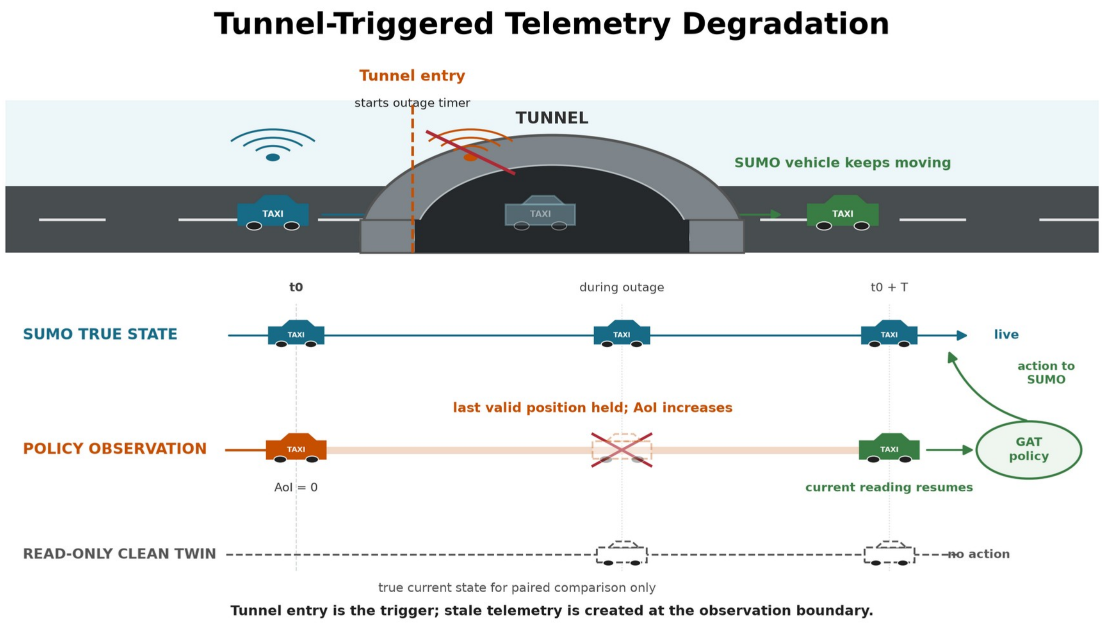

Figure 3.6. Tunnel entry triggers an observation-layer outage. SUMO keeps moving the vehicle while the policy sees its last valid position; the clean twin reads current state but never acts. The paired observations keep the same node identities, node order, masks, and selected action.

3.6 Explanation measures

3.6.1 Decision-level explanation faithfulness

DEF asks whether the nodes ranked highly by an explanation affect the chosen action more than a size- and type-matched random set. Two score functions are recorded from the same counterfactual forward passes. **Probability DEF** uses the selected action probability, p(G,a), and is bounded by the probability scale. **Logit-margin DEF** uses the selected action logit minus the strongest available alternative, m(G,a). The margin can retain resolution when the softmax probability is saturated, but its value depends on a checkpoint's logit scale.

Following ERASER (DeYoung et al., 2020), comprehensiveness measures output loss after removing top-k nodes R_k; sufficiency measures loss when only R_k remains. This study combines matched-random gains into DEF. The self node is always retained and is not a deletion candidate. The candidate set contains only valid peer-taxi and request nodes. The two gains are compared with five matched random subsets for each k in {1,2,3} and then averaged.

Let s(G,a) denote either p(G,a) or m(G,a). The action a is sampled from the policy observation being audited. For a degraded/clean pair, the action sampled on the degraded side is held fixed when both observations and every counterfactual subset are scored. For explanation subset R_k and each matched random subset indexed by b:

$$
\begin{aligned}
&\mathrm{Comp}(R_k,a)=\mathrm{s}(G,a)-\mathrm{s}(G\setminus R_k,a)\\
&\mathrm{Suff}(R_k,a)=\mathrm{s}(G,a)-\mathrm{s}(G[R_k],a)\\
&g_{\mathrm{comp}}(k;a)=\mathrm{Comp}(R_k,a)-\mathrm{mean}_b\,\mathrm{Comp}(B_k^b,a)\\
&g_{\mathrm{suff}}(k;a)=\mathrm{mean}_b\,\mathrm{Suff}(B_k^b,a)-\mathrm{Suff}(R_k,a)\\
&\mathrm{DEF}(a)=\mathrm{mean}_k\frac{1}{2}[g_{\mathrm{comp}}(k;a)+g_{\mathrm{suff}}(k;a)]
\end{aligned}
\tag{3.5}
$$

Comprehensiveness is better when removing the explanation weakens the action more than removing random nodes. Sufficiency is better when keeping the explanation loses less evidence than keeping random nodes. The logit margin is clipped to [-10, 10] only when an action becomes unavailable or unopposed; every clamp is logged.

Type-matched random subsets are the **control baseline**; they are not themselves a faithfulness score. Corrected DEF is the measured difference between the attention-ranked subset and this matched random control. Its sign is interpreted as follows:

1. **Positive DEF:** the attention-ranked nodes affect the selected action more than comparable random nodes. This supports decision-level faithfulness.
2. **DEF near zero:** the attention ranking has no measured advantage over the matched random control.
3. **Negative DEF:** the attention-ranked nodes affect the selected action less than the matched random nodes. This does not support the attention ranking as a faithful explanation.

These interpretations apply to both score functions. For no-op decisions, the primary metric uses standard type-matched DEF because no request action is selected. For dispatch decisions, it uses the chosen-action-protected variant so that the selected request cannot be removed. This hybrid action rule applies to both probability DEF and logit-margin DEF.

The predeclared H1-H4 analysis family and the paired clean-twin comparison use hybrid action-protected probability DEF, following the declared analysis plan. Logit-margin DEF is used for the construct-validity and ranking-control diagnostics because it can show output movement that a saturated probability hides. The two score functions have different units and cannot be converted into one another. Both use the type-matched baseline described in Section 3.7. DEF provides comparative perturbation evidence; even a positive value does not prove that attention is a complete causal explanation.

The study uses related measures for different questions. They should not be read as interchangeable estimates. Table 3.6 states the role of each measure.

Table 3.6: Explanation measures and roles

Measure	Decisions and score	Role in the study
Probability DEF (hybrid action rule)	standard type-matched score for no-op; selected request protected for dispatch	primary H1-H4 and paired clean/degraded analysis
Logit-margin DEF (hybrid action rule)	standard type-matched score for no-op; selected request protected for dispatch	construct-validity and ranking diagnostics when probability is saturated
Dispatch self-row margin-DEF	sampled dispatch actions only; declared self-row explanation	release check for decision relevance under clean and 60-second outage conditions
WAMSN	stale-exposed decisions; AoI-weighted attention	freshness-exposure indicator and H3 manipulation check
Paired attention and DEF shifts	degraded observation minus its clean twin	direct within-decision response to the observation-layer intervention

3.6.2 Stale-node attention

WAMSN measures attention attached to stale vehicle information:

$$\mathrm{WAMSN} = \frac{\Sigma_i\left(\mathrm{attention}_i \times \mathrm{normalized\_AoI}_i\right)}{\Sigma_i\left(\mathrm{attention}_i\right)} \tag{3.6}$$

Only self and peer-taxi nodes contribute because request nodes do not carry vehicle telemetry. The analysis reports WAMSN when at least one stale vehicle node is visible, together with stale-node attention share, stale-node mass, and whether a stale node enters the top three.

The paired stale-attention shift compares the degraded observation with its exact clean twin at the same simulation step. Positive values mean the degraded observation assigns more attention mass to nodes marked stale; negative values mean it assigns less.

3.7 Construct-validity audit

The audit supports the primary research problem by checking whether DEF is a valid comparison in this action-candidate graph.

Random masking without type matching can select passenger requests more often than the attention top-k. Removing such a request changes both the observation and the available action set. Masking a nearby taxi changes only the available information. The result can therefore reflect different node-type composition rather than explanation quality.

The corrected protocol uses three controls:

- **type matching:** random subsets contain the same mixture of taxi and request nodes as the explanation subset;
- **chosen-action protection:** the request associated with the selected action is not deleted in the action-protected variant;
- **clamp logging:** counterfactual action-margin clamps are counted so that action deletion can be detected.

Figure 3.7 illustrates why these controls are required.

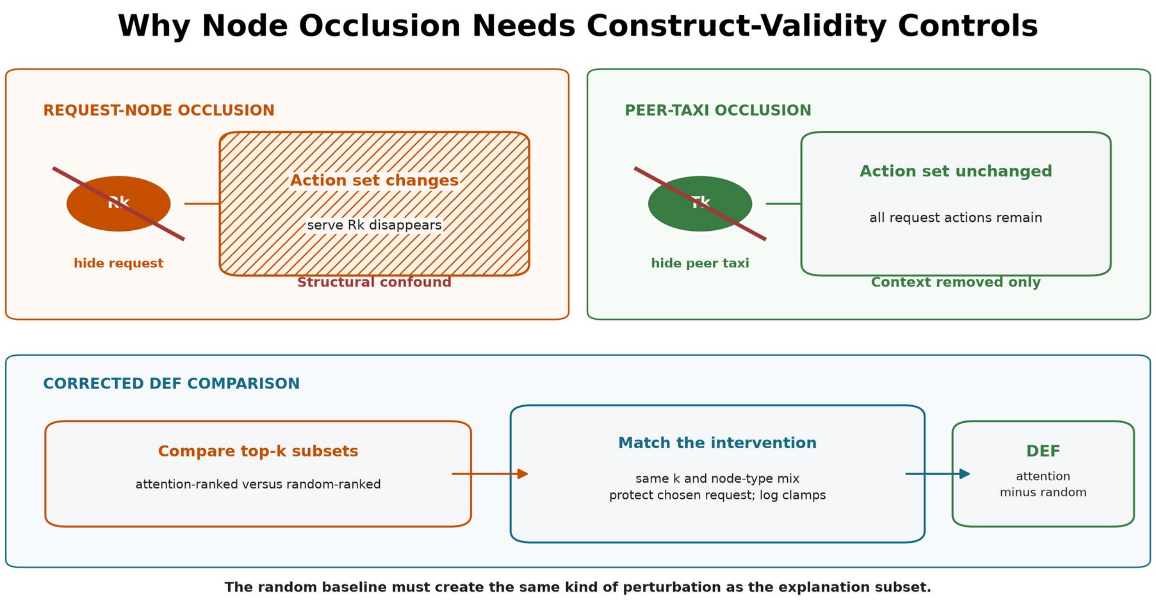

Figure 3.7. Request-node occlusion can remove its action, while peer-taxi occlusion removes context only. Corrected DEF matches subset size and node types, protects the chosen request, and logs margin clamps.

Uniform-baseline results are retained only as an audit trail. They are not used for the final hypothesis verdicts.

Three additional diagnostics test whether a near-zero DEF can be interpreted. First, a leave-one-out (LOO) control masks each valid node separately and ranks nodes by the resulting selected-action margin loss. The selected request is protected. This control is expected to strengthen comprehensiveness because it is built from the same single-node perturbation, but it is not an independent proof that the combined DEF is optimal. Second, Gradient x Input (GxI) ranks each node by the L2 norm of the selected-action logit gradient multiplied by its raw features. It is a post-hoc comparator that does not use attention weights. Third, the audit reports valid taxi/request counts and the exact expected top-k overlap with a type-matched random subset. Taxi-only Spearman correlation between attention and LOO effects provides a ranking measure that does not delete request actions and does not depend on random subset overlap.

3.8 Evaluation design

3.8.1 Primary evaluation

MLP, GAT, and GAT-Outage are evaluated under clean telemetry with eight evaluation seeds (42-49) and six episodes per seed. One checkpoint is frozen for each model and training seed, giving 48 held-out episodes per checkpoint. The clean capability evaluation contains 720 episode evaluations:

$$3 \times 5 \times 8 \times 6 = 720 \tag{3.7}$$

Table 3.7 summarizes the complete evaluation structure. An episode evaluation is one complete rollout of one frozen checkpoint under one telemetry condition.

Table 3.7: Held-out evaluation structure and episode counts

Evaluation	Model families	Total frozen checkpoints	Telemetry cases	Episodes/cell	Total
Clean capability context	3	15	1	48	720
Exposure-conditioned faithfulness sweep	2	10	5	48	2,400
Full ranking controls	2	10	2	9	180
Request-row sensitivity	2	10	2	9	180
Random-trigger sensitivity analysis	2	10	3	24	720

GAT and GAT-Outage receive the full faithfulness sweep:

$$2 \times 5 \times 5 \times 8 \times 6 = 2{,}400 \tag{3.8}$$

This gives 1,200 episode evaluations per GAT model and 2,400 across both models.

Table 3.8 lists the five telemetry conditions and their roles.

Table 3.8: Telemetry conditions used in held-out evaluation

Telemetry condition	Observation-layer treatment	GAT-Outage relation	Evaluation purpose
Clean	Current vehicle readings	No imposed outage	Capability context and faithfulness baseline
10 s	Freeze last valid reading for 10 seconds	Unseen shorter duration	Mild-degradation transfer test

Telemetry condition	Observation-layer treatment	GAT-Outage relation	Evaluation purpose
20 s	Freeze last valid reading for 20 seconds	Unseen shorter duration	Intermediate transfer test
30 s	Freeze last valid reading for 30 seconds	Training-matched duration	In-condition degradation test
60 s	Freeze last valid reading for 60 seconds	Unseen longer duration	Severe stress and transfer test

The legal-random capability baseline samples uniformly from no-op and the currently valid request actions. The greedy baseline is deliberately naive: each idle taxi chooses its nearest visible request independently. If several taxis choose the same request, the environment accepts the first action in the step and rejects the later conflicts; there is no fleet-wide assignment or conflict optimization. The policy is deterministic for a fixed demand variant, so its uncertainty is based on three independent demand variants rather than the repeated evaluation-seed records. It is therefore used as simple performance context, rather than as an optimized fleet-dispatch benchmark.

The relation column refers only to GAT-Outage; the clean-trained GAT has not seen any imposed outage duration during training. The evaluation changes the observation-layer condition but never retrains or reselects a checkpoint. Outside stale exposure, faithfulness is sampled every 16 decisions. Every decision that contains at least one stale vehicle node is scored. Each such observation is also compared with its exact clean twin at the same simulation step. The pair keeps the same trained checkpoint, agent, graph-node identities, node order, validity masks, and sampled action. Only telemetry-dependent feature values and freshness indicators differ. The degraded and clean evaluations also reuse the same matched-random subsets, so the paired difference does not contain avoidable baseline-sampling noise.

Each degraded condition must contain at least 20 stale-exposed records from at least three episodes. Preflight also checks expected cells, clean source provenance, type-matched controls, chosen-action exclusion, complete event capture, and separation in empirical AoI between conditions. A checkpoint sweep is included in the analysis only if it passes these gates.

3.8.2 Supporting analyses

Actions in the primary learned-policy evaluation are sampled from the policy distribution, matching training and preserving both no-op and dispatch decisions for audit. A separate descriptive diagnostic applies deterministic argmax to each of the ten GAT checkpoints over three held-out episodes. It does not affect checkpoint selection or the primary capability comparison.

The action-type diagnostic separates scorable decisions into no-op and dispatch; cases with no available request are excluded because no-op is then forced. Records are averaged by episode within each action group. This shows whether combined results apply to both action types or mainly reflect the more common action.

The random-trigger sensitivity analysis evaluates the ten frozen checkpoints under clean telemetry, a 30-second tunnel-triggered freeze, and a 30-second randomly triggered freeze. The per-taxi, per-step trigger probability is fixed at 0.0023 for every checkpoint. This value was chosen before the revised-model test, but the corrected policies follow different trajectories and produce less random-trigger exposure than tunnel-trigger exposure. The two conditions are therefore not treated as exposure matched. The analysis reports observed exposure and uses the random condition only to check whether the direction of the paired response depends entirely on the tunnel trigger. Its role is a supporting sensitivity analysis, not a confirmatory test.

The ranking controls use the same held-out demand and action-sampling protocol. They evaluate GAT and GAT-Outage under clean telemetry and a 60-second outage:

$$2 \times 5 \times 2 \times 3 \times 3 = 180 \tag{3.9}$$

The control evaluation uses seeds 42, 43, and 44 and covers all three held-out demand files in each run. LOO, Gradient x Input, overlap, and aggregation diagnostics are sampled every eight decisions. Query-row sensitivity uses the same cadence and retains only sampled selected-request actions. Records are averaged within episodes before bootstrap intervals are calculated. The pooled control CIs describe episode-level evaluation variation conditional on the five fixed checkpoints; they do not estimate uncertainty over a population of training runs. Per-training-seed control means are retained in the supporting data. Attention-aggregation sensitivity is reported by training seed for decisions with at least one visible stale vehicle node. These controls support interpretation of the main 48-episode-per-checkpoint sweep; they do not replace it.

3.9 Hypotheses and statistical analysis

Table 3.9 summarizes the five hypotheses, their analyses, and their roles. H1, H3, and H4 were predeclared as primary tests. H2 is a supporting exploratory diagnostic rather than a test of the main research claim, and H5 is a matched-seed descriptive comparison.

Table 3.9: Hypothesis map and analysis roles

Hypothesis	Question	Analysis	Role
H1	DEF across outage durations	episode-block probability-DEF trend	primary
H2	Faithfulness versus capability decline	paired clean-relative change	supporting exploratory diagnostic
H3	WAMSN across outage durations	episode-block WAMSN trend	primary check
H4	Within-episode WAMSN-DEF link	Spearman correlation and sign-flip test	primary
H5	Weaker relationships for GAT-Outage	matched-seed correlation comparison	descriptive

RQ1 uses clean DEF, type-matched random controls, LOO, and ranking controls. RQ2 uses the paired clean/degraded stale-attention comparison. RQ3 uses H1, H3, and H4 together with the paired and action-type diagnostics. RQ4 uses H5 to describe association strength, together with the per-checkpoint direction and decision-relevance results to assess consistency. Weaker absolute correlation is not itself evidence of lower variability or greater robustness.

3.9.1 Statistical procedure

H1 and H3 use episode-block permutation trend tests and cluster bootstrap confidence intervals. H2 uses paired cell-level sign flips relative to each evaluation seed's clean condition. H4 first calculates a Spearman correlation within each episode and then applies a sign-flip test. Holm adjustment is applied to H1-H4 within each trained policy; H2 remains exploratory.

All tests use 10,000 permutation draws, 2,000 bootstrap draws, statistical seed 0, and 95% percentile bootstrap intervals. Episodes, rather than individual decisions, are the resampling blocks. Each training seed represents one trained policy. Five seeds can show variation but provide limited support for a population-level claim, so cross-seed results are reported as ranges and counts of supporting seeds without a population p-value.

H5 is supported for a matched seed only when both the outage-duration/DEF and WAMSN/DEF correlations are closer to zero for GAT-Outage than for GAT. The result is the number of supporting pairs out of five; no cross-seed significance test is used. This criterion measures weaker association, not stability across training seeds or causal improvement from outage training.

3.9.2 Interpretation limits

H2's original rate divides by clean DEF. Because clean DEF is close to zero, that ratio can exaggerate a small absolute change. The analysis therefore uses the support verdict and absolute trajectories as its main evidence.

H3 is a manipulation check, not evidence that AoI causes DEF to change. WAMSN contains normalized AoI by definition, so H3 only confirms that longer outages create more AoI-weighted stale exposure.

H5 uses matched training budgets and learning-rate schedules. However, validation telemetry still matches each model's training condition, so the comparison does not isolate training observations from checkpoint selection. The exposure-conditioned paired follow-up is also kept separate from H1-H5: it subtracts clean-twin DEF from degraded DEF for the same decision and reports episode-block intervals, effect size, direction across trained policies, and uncertainty.

3.10 Audit decision implementation

The experiment pipeline produces measurements; a separate release-audit stage turns those measurements into a decision about whether attention may be shown as an explanation. The stage reads saved evidence and leaves the policy, selected action, SUMO state, and observations unchanged.

Its inputs are the frozen-checkpoint preflight reports, condition summaries, action-type results, training-seed synthesis, deterministic diagnostics, faithfulness controls, and tunnel-versus-random trigger comparison. These artifacts jointly cover telemetry freshness, decision relevance, action composition, attention extraction, checkpoint replication, and the action rule intended for deployment.

The implementation applies nine checks:

1. evidence integrity;
2. separate freshness reporting;
3. type-matched and action-protected controls;
4. dispatch-action decision relevance;
5. no-op and dispatch composition;
6. attention-extraction stability;
7. consistency across independently trained checkpoints;
8. capability of the action rule intended for deployment;
9. robustness to tunnel and random loss triggers.

Decision relevance passes only when the 95% confidence-interval lower bound of self-row dispatch margin-DEF is greater than zero under both clean and 60-second outage evaluation. A positive value means that the attention ranking is more informative than its type-matched random control. Extraction stability fails if an audited layer, head, maximum, or rollout rule reverses the direction produced by the declared self-row mean. Trigger robustness uses confidence intervals: a direction is supported only when its 95% interval excludes zero, and supported tunnel and random-trigger directions must agree. An interval that crosses zero is INDETERMINATE, not evidence of a reversal. The remaining rules test evidence availability, action coverage, checkpoint agreement, and the action rule declared for deployment.

The present audit concerns actions sampled from the policy distribution. The argmax evaluation is therefore retained as a descriptive limitation rather than a mandatory release gate. If an argmax policy were intended for deployment, its held-out capability check would become mandatory. The current action-composition gate requires both action groups and at least 20 dispatch decisions at the model and individual-checkpoint levels, matching the minimum stale-exposure record floor used by the evaluation preflight. It confirms that dispatch evidence is present but does not by itself establish that a sparse dispatch group is representative.

The decision concerns the whole declared self-row explanation channel. Its dispatch criterion is a necessary gate; failure does not establish that every no-op explanation is unfaithful. Section 4.12 separately reports an exploratory absolute no-op analysis from archived records.

The output has three states. ELIGIBLE means that every required check passed within the stated scope. WITHHOLD means that at least one check failed and attention must remain an internal diagnostic. INCOMPLETE means that required evidence is missing, so no release decision can be made. Eligibility is not proof of a complete causal explanation; it only permits presentation with an explicit freshness indicator and audit scope as an audited candidate explanation.

The release rules were developed from an earlier pilot and fixed before the present rerun. Their application is separate from the H1-H5 tests. Because the rerun uses the same scenario and demand splits as the pilot, it does not establish transfer to an independently collected dataset. Future evaluations should retain the rules and thresholds before examining test results.

3.11 Implementation and reproducibility

3.11.1 Software environment and provenance

The experiment runner records the configuration, commands, seeds, checkpoint hashes, source revision, and preflight reports in versioned run directories. Compact outputs include the summary, performance context, deterministic diagnostic, action-type audit, and training-seed synthesis tables. The training-seed synthesis makes consistency across trained policies explicit. Reproduction commands, file locations, and expected outputs are documented in docs/REPRODUCE_EXPERIMENTS.md. The complete frozen-checkpoint workflow is available as a single framework stage. It produces a machine-readable JSON decision, a CSV check table, and a reader-facing Markdown report. The detailed input and output contract is documented in docs/FRESHNESS_AWARE_EXPLANATION_AUDIT.md.

All reported models were trained and evaluated on one Apple M3 Pro workstation with 12 logical CPU cores, 18 GiB of installed memory and macOS 14.2.1. The runtime uses Python

3.11.14, NumPy 1.26.4, PyTorch 2.2.2, Gymnasium 1.3.0, PettingZoo 1.26.1 and SUMO, TraCI and sumolib 1.20.0. Training and evaluation use CPU execution, with one Torch intra-operation thread and the default 12 inter-operation threads. SUMO runs through TraCI. Training, selection and clean performance evaluation run sequentially. The main sweep, random-loss checks and attribution controls use up to six independent evaluation processes, each with a separate output directory and the same per-process settings. Execution records retain worker commands and timings; the overlapping batch is reported separately from sequential stages.

The SUMO command-line binary was built from the unmodified upstream 1.20.0 source tag using AppleClang 15.0.0, Xerces-C 3.3.0 and PROJ 9.7.0. The source revision, binary hash, exact Python dependency lock and build commands are retained with the experiment records. Network generation remains a separate step using the version recorded in Section 3.2.

Training all 15 models took 491.8 seconds at the orchestration-stage level, including process startup and checkpoint output. Checkpoint selection took 633.6 seconds, clean performance evaluation 585.7 seconds, and the deterministic diagnostic 34.9 seconds. The overlapping main-sweep, random-loss and attribution-control batch took 1,684.7 seconds at the wrapper level. Later robustness and control stages reused the completed cells, so their short durations are aggregation overhead rather than simulation cost. Main hypothesis analysis took 232.5 seconds and baseline evaluation 65.4 seconds. The execution records retain the remaining analysis and packaging durations. These are measurements on this workstation, not comparisons with the earlier machine, whose hardware and complete timings were not recorded.

The initial SUMO 1.27.0 attempt stopped on a reservation-data decoding error. Its partial and completed runs were excluded, and all models restarted under the fixed 1.20.0 runtime in a fresh output directory. No result from that attempt or from the earlier workstation enters the reported estimates.

3.11.2 Component interfaces and feature configuration

Table 3.10 connects the components to their implementation and explains why each is included. Package modules below are within dispatch_marl; the standalone analysis entry point is under scripts. The simulator advances physical state independently of the observation cache, which is necessary to construct the clean twin without changing the world.

Table 3.10: Implementation components and design rationale

Component	Implementation interface	Purpose and design rationale
Simulator and local graph	env.py; scenario.py	Controls taxi dispatch and builds typed observations from a reproducible road and demand scenario
Telemetry cache	degradation.py	Freezes received readings while SUMO continues moving; separates stale evidence from ground truth
Typed encoder and actor	models/policy.py; models/gat.py	Projects heterogeneous features, combines local context and scores valid request actions
Centralized critic and PPO	models/policy.py; training.py	Estimates team value during training; preserves local actor execution and actual decision intervals

Component	Implementation interface	Purpose and design rationale
Faithfulness evaluator	faithfulness.py	Compares ranked nodes with matched random subsets while protecting the selected action
Statistical analysis	scripts/analyze_hypotheses.py	Aggregates by episode, estimates uncertainty and preserves per-checkpoint variation
Release audit	explanation_audit.py	Applies explicit evidence checks and produces a traceable release decision

Each node type has five numeric features. Self features are normalized x and y position, episode time, speed and AoI. Each of the five peer slots holds relative x/y, empty status, distance and AoI; each of the five request slots holds pickup x/y and drop-off x/y relative to the acting taxi, plus waiting time. Self position uses the network bounding box; relative positions and distances use its diagonal. Time uses the 1,200-second episode horizon. Speed, AoI and waiting time use scales of 30 m/s, 60 seconds and 600 seconds, respectively, with their upper normalized values clipped to one. The observation also includes peer and request validity masks and a separate position-valid flag.

The actor has up to 11 nodes, each projected to width 64, two attention layers and four heads per layer. The attention tensor for one observation therefore has two layers of four 11-by-11 matrices before padded entries are excluded. The six action slots are no-op and five request candidates. Inference and audit use evaluation mode; a sampled selected action is held fixed when its clean twin and counterfactual observations are scored. Neither a fresh twin nor a masked input triggers a new action sample for that comparison.

3.11.3 End-to-end algorithm

The following pseudocode summarizes the implemented dependency order. It separates training and validation from the held-out evaluation and the post-study synthesis of release rules.

```
1. Load the fixed road network and the train/validation/test manifest.
2. For each model and training seed:
   train on training demand using interval-aware PPO updates;
   select a trained checkpoint using validation completed journeys;
   freeze the checkpoint and save its hash and configuration.
3. For each frozen checkpoint and held-out telemetry condition:
   roll out the sampled-action actor in SUMO;
   capture every stale-exposed decision and scheduled clean samples;
   construct its clean twin with the same nodes, masks and action;
   extract the declared attention ranking;
   score ranked and type-matched random subsets with action protection;
   reuse random subsets across each clean/degraded pair;
   save scores, freshness, action type and episode identity.
4. Check evidence completeness; aggregate and resample by episode.
5. Run ranking, extraction, action-type and trigger diagnostics.
6. Apply the stated audit rules; emit decisions and failed-check reasons.
```

The minimal baseline and full audit commands are documented in the reproduction guide. The added EDA is reproduced separately with python scripts/plot_dataset_eda.py; its JSON summary records source hashes and the distance definition. It does not alter checkpoints or confirmatory hypothesis outputs.

3.12 Ethics and data governance

The experiment uses a simulated taxi fleet, generated demand, and an OpenStreetMap-derived road network. It contains no human participants, personal records, live vehicle identifiers, or commercial dispatch data. Its main ethical risk is miscommunication: presenting attention as a trustworthy reason could give an operator false confidence. The reporting therefore keeps freshness, task behavior, and faithfulness separate and avoids unsupported causal claims.

Chapter 4: Results

All 15 training runs passed the stability gates, all 120 clean evaluation cells completed, and all ten faithfulness sweeps passed their preflight checks. Results are reported by training seed because one checkpoint, rather than each decision, is the independent trained-policy replicate. Unless stated otherwise, intervals are 95% confidence intervals (CIs) formed from episode blocks.

4.1 Policy capability and training stability

Validation selected different epochs for different training seeds, as shown in Table 3.3. Epoch 0, saved after the first training epoch, was retained only as an early-training diagnostic and could not be selected. Every training run passed the declared stability gates.

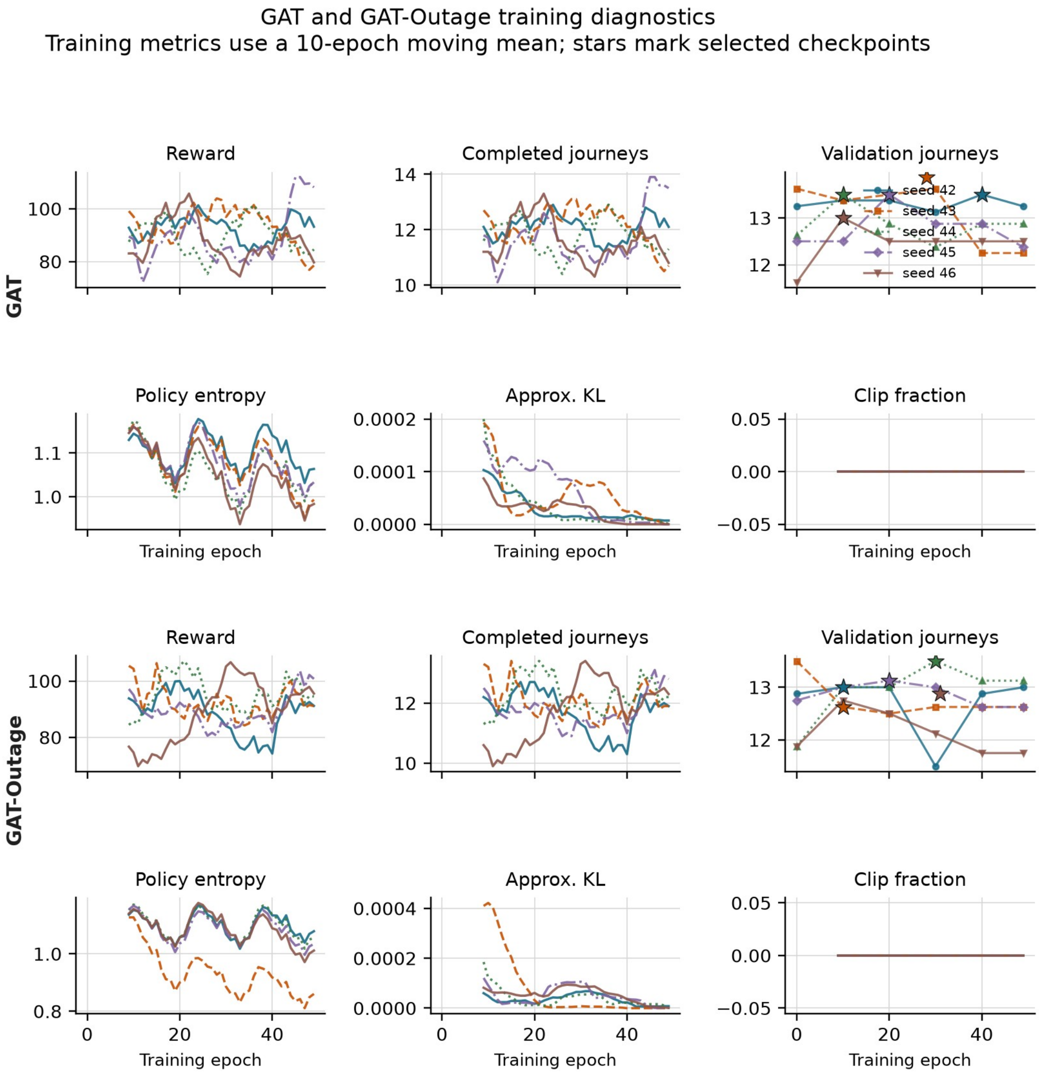

Figure 4.1. Training diagnostics for GAT (upper panels) and GAT-Outage (lower panels). Curves use a ten-epoch moving mean; stars mark the checkpoints selected on validation data. All ten GAT runs passed the stability gates, although the curves still show variation between epochs.

Table 4.1 reports clean held-out capability. The learned policies complete between 11.729 and 13.146 passenger journeys per episode across individual training seeds. Their results overlap the legal-random CI and the range of the simple greedy policy. Therefore, the comparison establishes that the audited policies can dispatch and complete journeys; it does not show that GAT is better than the baselines or MLP.

Table 4.1: Policy capability on held-out demand

Policy	Mean completed journeys per episode	Uncertainty or training-seed range
Legal random	12.708	95% episode CI [11.875, 13.542]
Greedy nearest request	12.000	95% demand-variant CI [10.000, 15.000]
MLP	12.521	training-seed range 11.729-13.146
GAT	12.850	training-seed range 12.792-12.917
GAT-Outage	12.721	training-seed range 12.271-12.958

The legal-random estimate uses 48 episode records. The greedy policy is deterministic for a fixed demand file, so its effective replication unit is the three held-out demand variants. Each learned-policy point uses 48 held-out episodes: eight action-sampling seeds and six episodes per seed.

The lower greedy mean may partly reflect independent taxis selecting the same request without coordination. However, the baseline records do not quantify rejected conflicts, so their contribution to the difference is unknown. The point estimates alone do not establish that random dispatch is generally better than greedy dispatch.

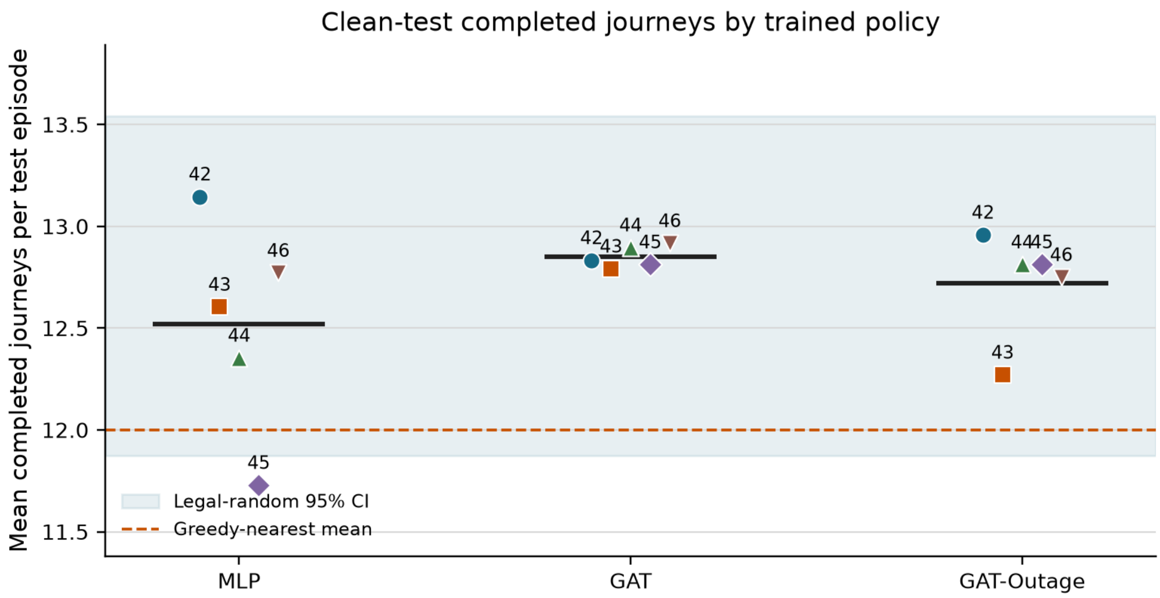

Figure 4.2. Clean-test completed journeys for each selected policy. The shaded band is the legal-random 95% CI and the dashed line is the greedy mean. Performance shows whether the policies can complete journeys; explanation faithfulness is evaluated separately.

The separate argmax diagnostic also completes journeys. Across three held-out episodes, GAT checkpoint means are 7.3, 7.7, 8.7, 11.3 and 5.3 for seeds 42-46; GAT-Outage means are 9.0, 7.7, 10.0, 9.0 and 8.7. None of the 30 checkpoint-episodes has zero completed journeys.

The argmax means are lower than the corresponding sampled-action means. This suggests that the action-selection rule matters, but the diagnostic uses only three episodes per checkpoint, compared with 48 for sampling. A deployment using argmax would need its own matched evaluation; the present capability claim concerns the sampled policies.

4.2 Protocol correction and precision audit

The corrected experiment addresses four implementation risks found during the code review: rewards earned while a taxi was busy are now accumulated over the whole decision interval; completed passenger journeys are counted globally; request-linked actions are protected in dispatch DEF; and unrounded numeric values are stored for analysis. All models were trained afresh under the corrected protocol, followed by validation selection and test evaluation on the current workstation. No result is combined with an earlier run.

Earlier pilot estimates are superseded because they were obtained under a protocol with known implementation defects. The rerun also changes the trained checkpoints, seed coverage, training budgets and runtime environment. Differences from the pilot cannot therefore be attributed to an individual correction, or used to estimate the effect of any one implementation change.

Table 4.2 reports the numeric-precision check. Repeating the hypothesis analysis after rounding stored values to four decimals preserved the signs of H1 rho, H4 rho and paired DEF for all ten checkpoints, but changed one H1 verdict. Full precision remains the official analysis.

Table 4.2: Protocol correction and numeric-precision audit

Precision check	Checkpoints tested	Point-estimate sign changes	H1-H4 significance decisions changed	Use in thesis
Full precision versus four-decimal sensitivity	10	0	1 (H1, GAT-Outage seed 46)	full precision

Sign changes refer to H1 rho, H4 rho and the paired DEF point estimate. The next column counts H1-H4 tests crossing the Holm-adjusted 0.05 threshold. Changes in whether the paired DEF confidence interval includes zero are reported separately below.

For GAT-Outage seed 46, the H1 Holm-adjusted p-value changes from 0.0495 to 0.0867 after rounding. Its paired DEF interval also changes from crossing zero to slightly above zero. This borderline checkpoint is precision-sensitive and should not carry a general claim. The full-precision result remains official, while the rounding check qualifies its interpretation.

4.3 Decision-relevance controls

The supporting control experiment evaluates raw attention, Gradient x Input (GxI), and leave-one-out (LOO) rankings under clean and 60-second outages. Each model-condition estimate combines 45 episode blocks: five training seeds, three evaluation seeds, and three episodes.

Table 4.3: Decision-relevance controls under clean and degraded telemetry

Model	Condition	Raw-attention margin-DEF [95% CI]	GxI margin-DEF [95% CI]	LOO margin-DEF [95% CI]
GAT	clean	-0.000907 [-0.001075, -0.000729]	-0.000366 [-0.000986, +0.000161]	+0.003228 [+0.002591, +0.003927]
GAT	60 s	-0.000767 [-0.000916, -0.000612]	-0.000014 [-0.000461, +0.000378]	+0.003066 [+0.002492, +0.003698]
GAT-Outage	clean	-0.002493 [-0.003356, -0.001674]	+0.000131 [-0.000671, +0.000958]	+0.004328 [+0.003486, +0.005286]
GAT-Outage	60 s	-0.002307 [-0.003045, -0.001613]	+0.000389 [-0.000367, +0.001149]	+0.003941 [+0.003194, +0.004787]

Raw attention does not beat its type-matched random control: all four CIs are below zero. LOO is positive in every case. This shows that the evaluator can reward a perturbation-based ranking, but LOO is a positive control rather than a ground-truth explanation. All four GxI intervals include zero.

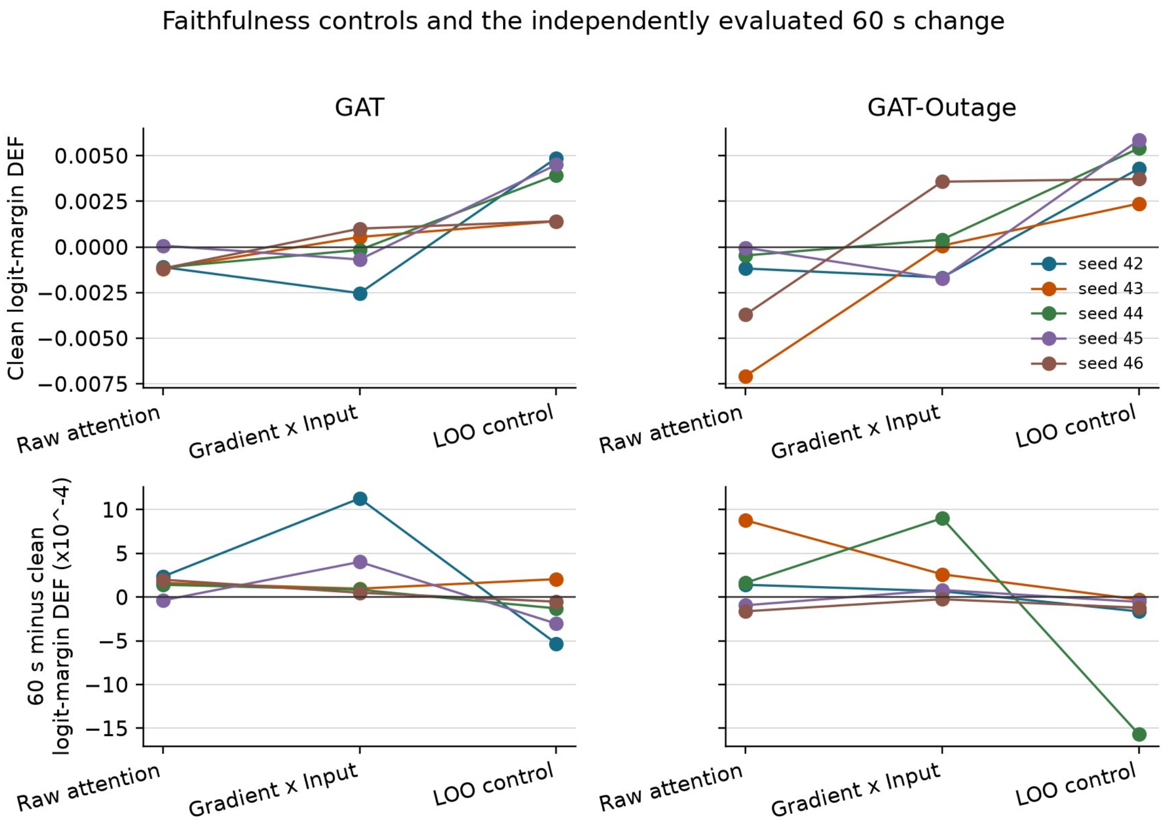

Figure 4.3. Clean logit-margin DEF (top) and independently evaluated 60-second minus clean change (bottom), by training seed. Pooled raw attention shows no matched-random advantage.

The attention result also depends on how weights are extracted. Every checkpoint has at least one audited layer, head, maximum, or rollout choice that reverses the direction of its default stale-attention shift.

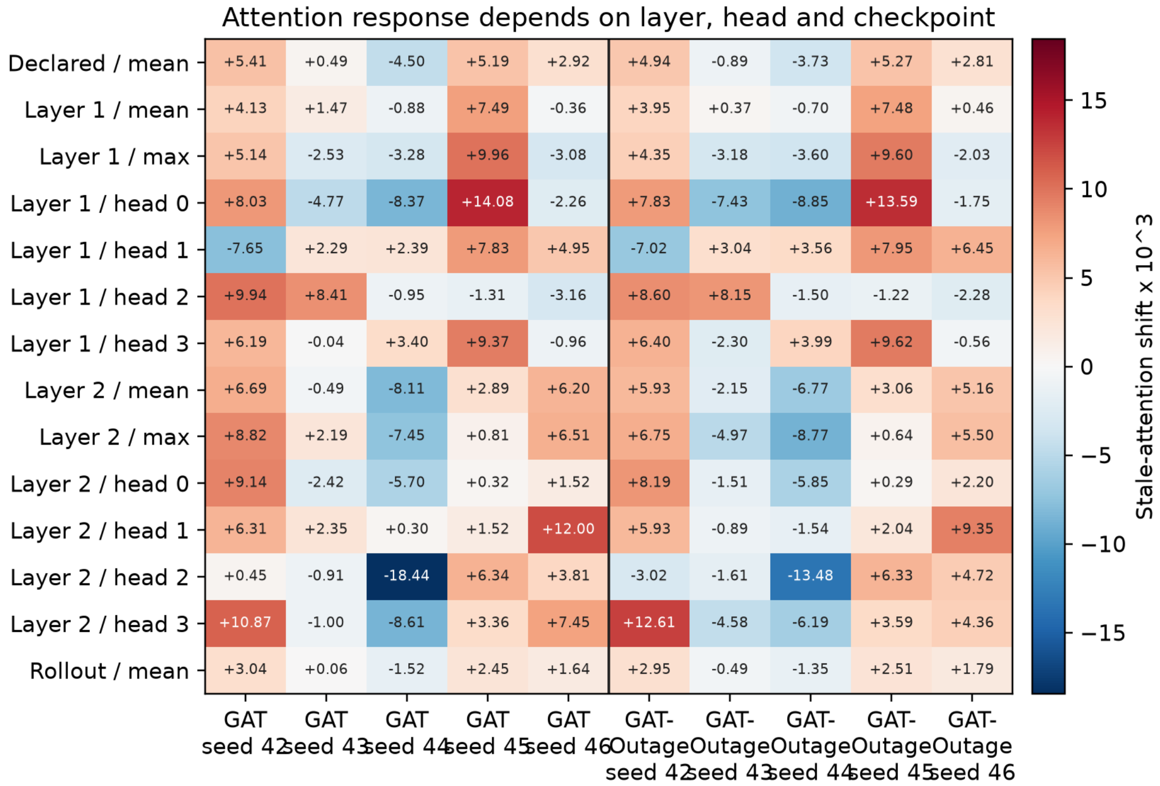

Figure 4.4. Stale-attention shift under alternative extraction rules, multiplied by 1,000. Positive and negative values within the same checkpoint show that the conclusion can depend on the chosen layer or head.

For dispatch decisions, the declared self-row margin-DEF is negative for both models under clean and 60-second conditions. Using the selected request row gives a different result. GAT remains below zero in both conditions, while GAT-Outage is positive: +0.000711 [0.000252, 0.001219] when clean and +0.000762 [0.000297, 0.001254] at 60 seconds. This is positive evidence for the alternative row in this configuration, but it does not validate the predeclared self-row channel or establish consistency across both model conditions.

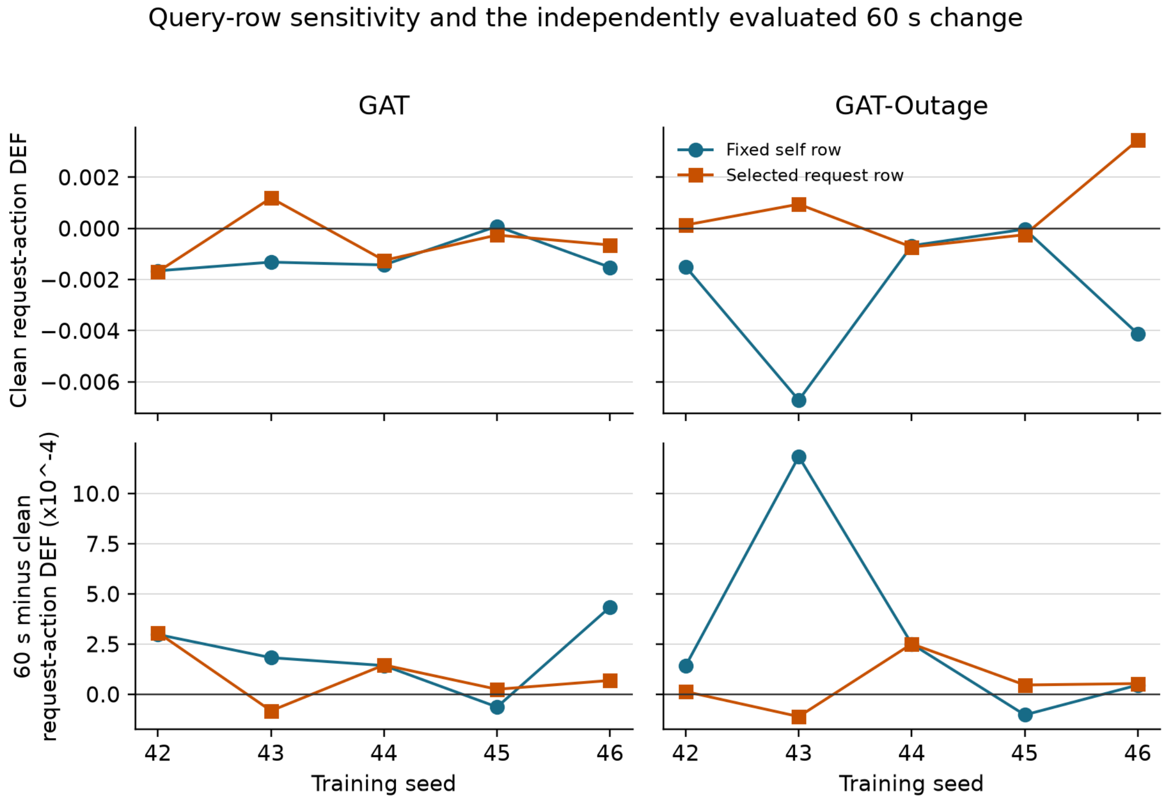

Figure 4.5. Dispatch-action margin-DEF from the fixed self row and selected request row. The lower panels show the independently evaluated 60-second minus clean change. The selected request row improves GAT-Outage estimates above zero, while GAT remains below zero. The result is specific to the extraction rule and model configuration.

4.4 Small-graph resolution

The scored graph contains five visible peer taxis in nearly every record and a smaller, variable number of request nodes. Table 4.4 describes the eligible non-self nodes in main-sweep records with at least one available request and a recorded primary DEF. The distribution is descriptive and weighted by scored decisions.

Table 4.4: Valid node-count distribution in scored graphs

Model	Node type	Mean	Median	5th percentile	95th percentile
GAT	peer taxis	5.00	5	5	5
GAT	requests	2.54	2	1	5
GAT	all non-self nodes	7.54	7	6	10
GAT-Outage	peer taxis	5.00	5	5	5
GAT-Outage	requests	2.54	2	1	5
GAT-Outage	all non-self nodes	7.54	7	6	10

In a small candidate set, a top-k explanation and a matched random set can overlap by chance. For both models, attention-LOO agreement is below the expected random overlap at k=1, 2, and 3. The overlap reduces DEF's ability to separate rankings, especially as k increases.

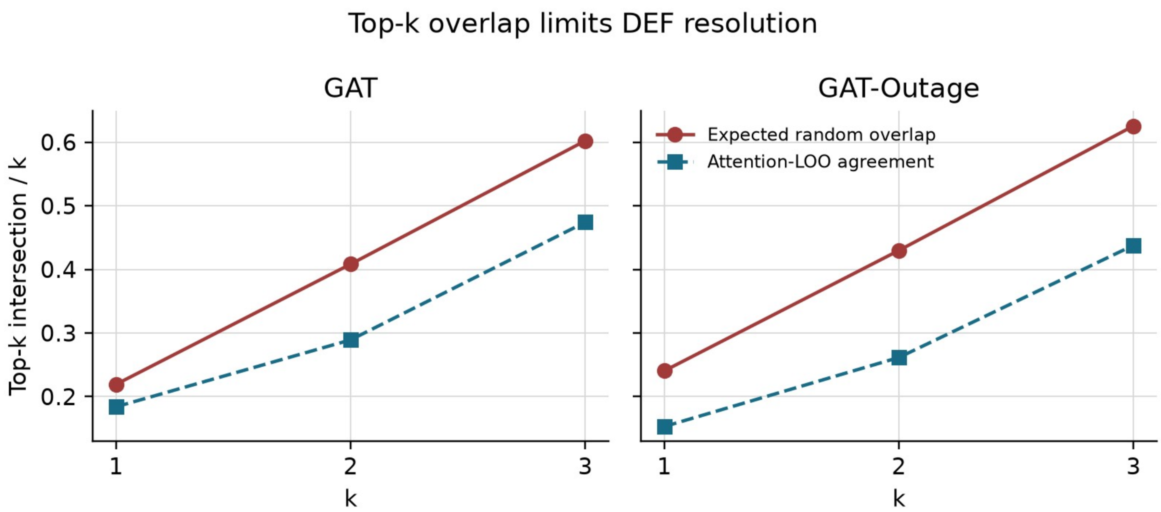

Figure 4.6. Observed attention-LOO agreement compared with expected random overlap. Large expected overlap limits the resolution of a top-k comparison; it does not by itself prove or disprove faithfulness.

4.5 Telemetry manipulation and stale exposure

The outage manipulation worked as intended. Conditional WAMSN rises from 0.027-0.029 at 10 seconds to 0.071-0.077 at 60 seconds. Across the ten checkpoints, H3 correlations between duration and WAMSN are 0.466-0.486 and remain significant after Holm correction. H3 is therefore supported for all ten checkpoints.

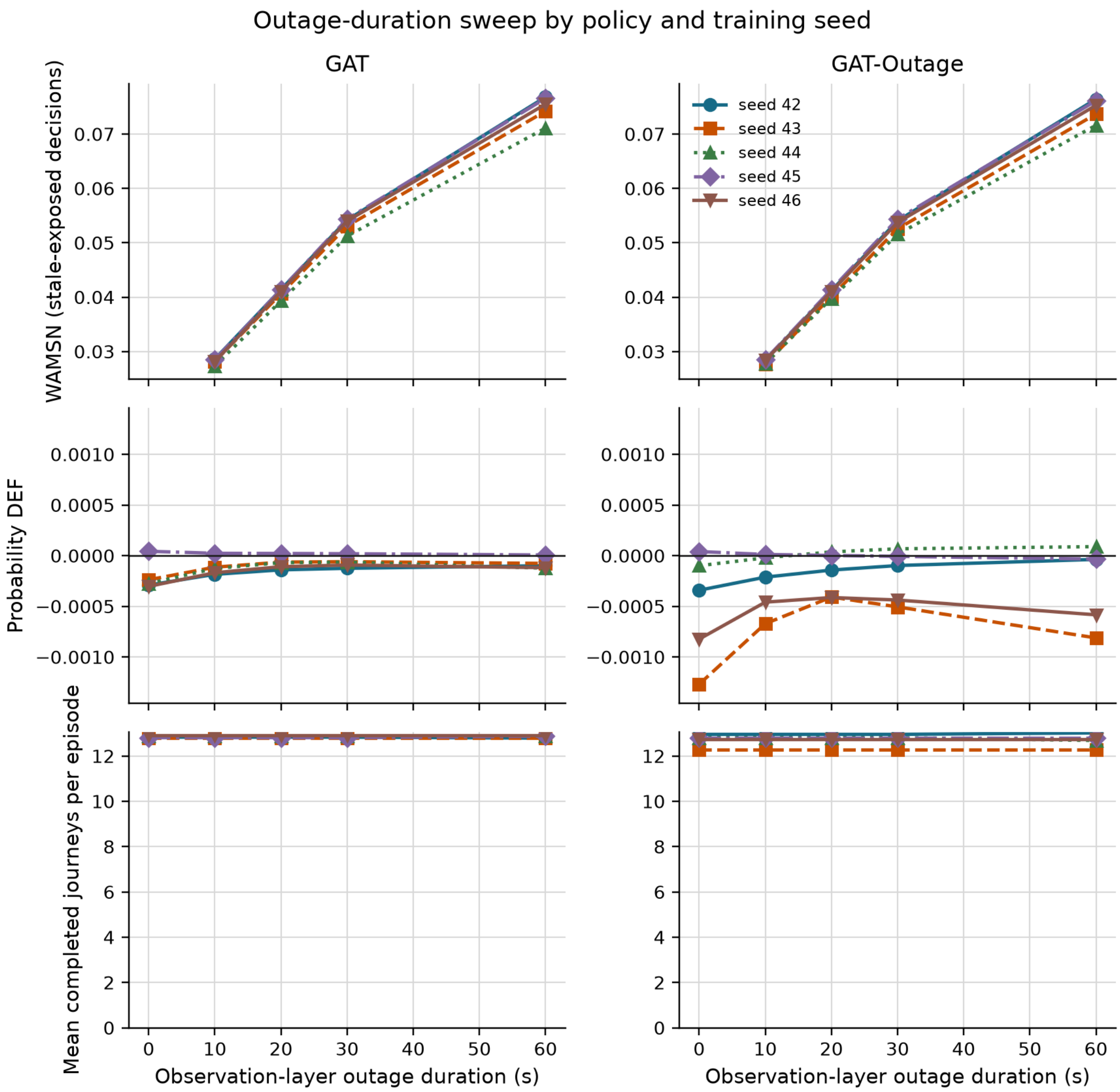

Figure 4.7. Longer observation outages increase WAMSN for every checkpoint. Completed journeys remain broadly stable. Probability DEF has no consistent direction across checkpoints. WAMSN verifies stale exposure; it does not show that AoI causes a change in faithfulness.

4.6 Paired clean and degraded observations

The paired audit compares each degraded observation with its exact clean twin at the same SUMO step. Positive attention shift means that the degraded observation assigns more attention to nodes marked stale. Positive DEF shift means that measured DEF is higher, not lower, under degradation.

Table 4.5: Exposure-conditioned paired attention and DEF shifts

Model	Seed	Stale-attention shift x 1,000 [95% CI]	Probability-DEF shift x 10,000 [95% CI]
GAT	42	+3.805 [+3.689, +3.930]	+0.653 [+0.460, +0.872]
GAT	43	+0.650 [+0.636, +0.665]	+1.225 [+1.097, +1.342]
GAT	44	-3.444 [-3.530, -3.354]	+0.819 [+0.747, +0.891]
GAT	45	+4.022 [+3.929, +4.116]	-0.075 [-0.104, -0.048]
GAT	46	+2.474 [+2.418, +2.527]	+0.828 [+0.720, +0.940]
GAT-Outage	42	+3.331 [+3.233, +3.431]	+0.965 [+0.840, +1.095]
GAT-Outage	43	-0.736 [-0.769, -0.702]	+1.344 [+1.176, +1.514]
GAT-Outage	44	-2.440 [-2.512, -2.367]	+1.619 [+1.459, +1.774]
GAT-Outage	45	+3.918 [+3.823, +4.009]	-0.445 [-0.494, -0.398]
GAT-Outage	46	+2.083 [+2.033, +2.132]	+0.039 [-0.019, +0.103]

The estimates summarize stale-exposed decisions across the evaluated degraded conditions using the saved episode-block analysis. Attention uses all eligible stale-exposed records; DEF additionally requires a scorable action, so their record and episode-block counts can differ. They do not estimate variation across new, independently trained policies. The unscaled paired probability-DEF changes range from approximately -0.000044 to +0.000162. These are changes in the composite, random-adjusted DEF score, not percentage-point changes in success probability. Seven checkpoints have positive intervals, both seed-45 checkpoints have negative intervals, and GAT-Outage seed 46 is inconclusive. No deployment utility threshold for these small effects has been validated.

Seven checkpoints show a clear increase in attention to stale nodes and three show a clear decrease: GAT seed 44 and GAT-Outage seeds 43 and 44. The two models share the direction for four of five matched seeds. This pattern shows substantial checkpoint dependence, rather than a uniform effect of outage training.

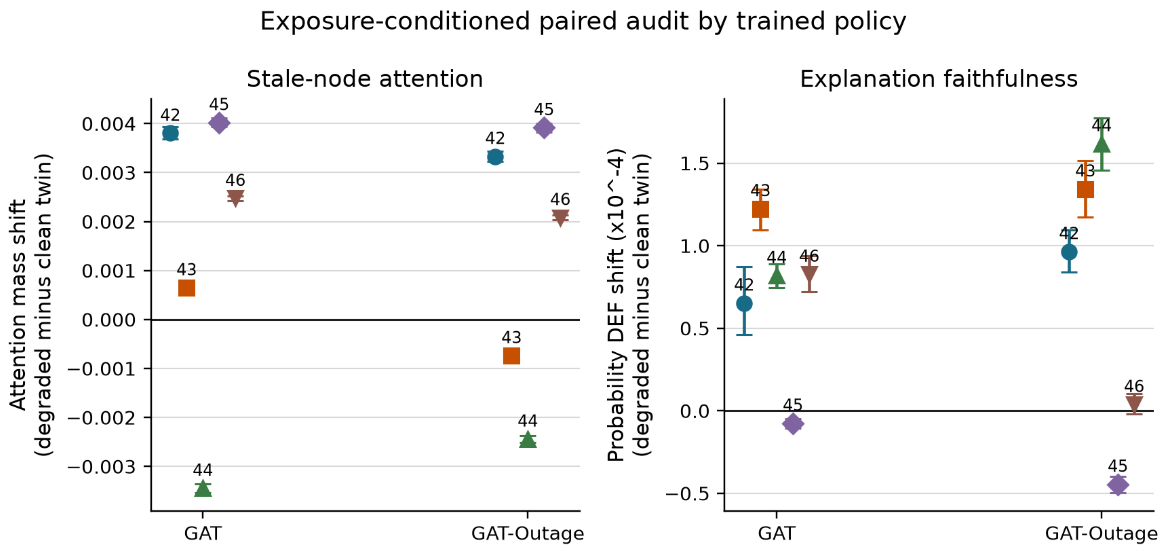

Figure 4.8. Exposure-conditioned degraded-minus-clean shifts by frozen checkpoint. Error bars are episode-block bootstrap CIs. Both attention and probability DEF vary across checkpoints. Seven DEF intervals are positive, two negative, and one crosses zero.

4.7 Analysis by action type

Dispatch accounts for 14,953 of 20,096 scorable GAT decisions (74.4%) and 15,714 of 20,056 GAT-Outage decisions (78.3%). No-op accounts for the remaining 5,143 and 4,342 records. Table 4.6 separates the paired changes by checkpoint and action type.

Table 4.6: Action composition and paired DEF by action type

Checkpoint	No-op (%)	Dispatch (%)	All-action DEF shift x 10,000	No-op DEF shift x 10,000	Dispatch DEF shift x 10,000
GAT, seed 42	29.8	70.2	+0.653	-0.193	+1.041
GAT, seed 43	21.5	78.5	+1.225	-2.483	+2.345
GAT, seed 44	24.3	75.7	+0.819	-0.630	+1.399
GAT, seed 45	25.5	74.5	-0.075	-0.010	-0.071
GAT, seed 46	26.9	73.1	+0.828	-0.167	+1.443
GAT-Outage, seed 42	25.7	74.3	+0.965	-0.416	+1.540
GAT-Outage, seed 43	7.3	92.7	+1.344	-0.644	+1.566
GAT-Outage, seed 44	31.6	68.4	+1.619	-1.619	+3.537
GAT-Outage, seed 45	23.0	77.0	-0.445	+0.157	-0.646
GAT-Outage, seed 46	20.6	79.4	+0.039	-0.065	+0.065

No-op DEF decreases in all five GAT checkpoints and four of five GAT-Outage checkpoints. Dispatch DEF increases in eight checkpoints and decreases in both seed-45 checkpoints. Nine checkpoints therefore have opposite no-op and dispatch directions. GAT seed 45 decreases in both groups; GAT-Outage seed 45 increases for no-op but decreases for dispatch. The combined estimate does not describe every action type equally well.

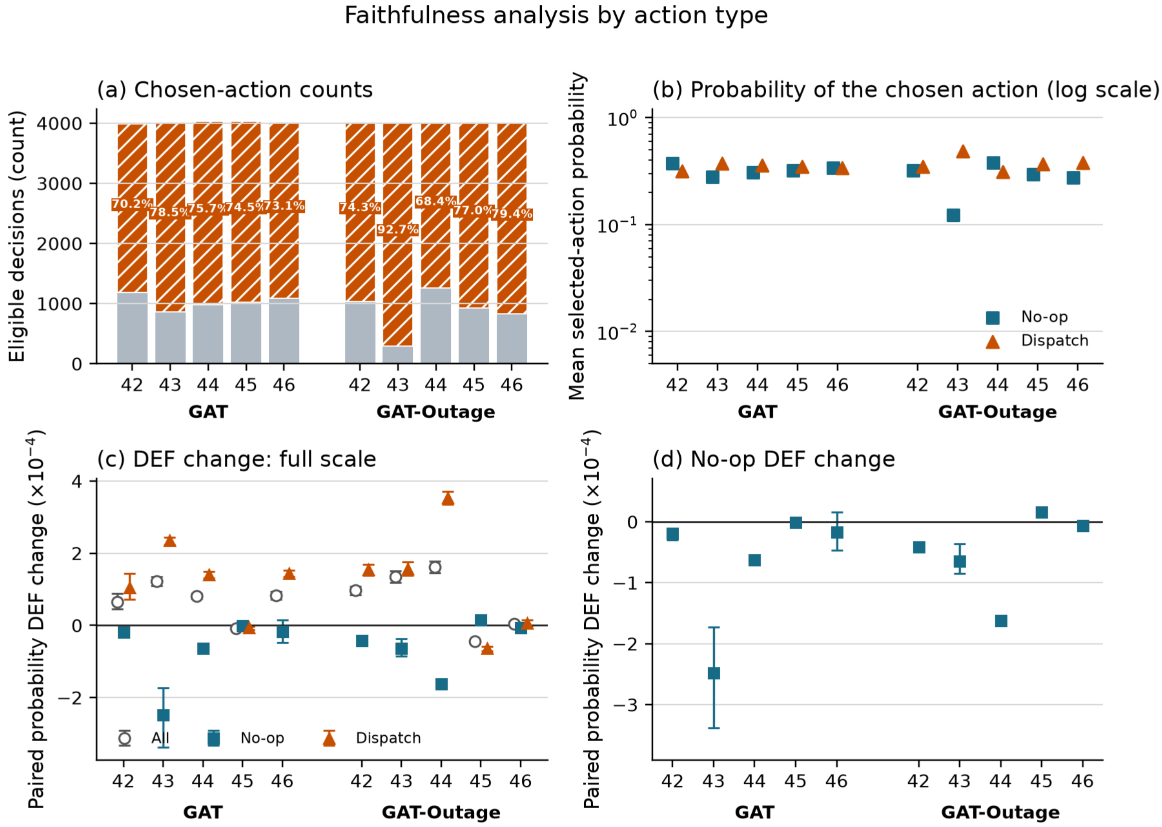

Figure 4.9. (a) Scorable action counts and dispatch share; (b) selected action probability; (c) all, no-op, and dispatch paired DEF changes; and (d) an enlarged view of the no-op changes. Error bars are episode-block bootstrap CIs.

4.8 Random-trigger sensitivity analysis

The random-loss control replaces tunnel entry with an independent per-taxi, per-step trigger while retaining a 30-second observation-layer freeze. The configured trigger rate is fixed, but observed random exposure is only about 15% of tunnel exposure (median across checkpoints). This is a trigger-location sensitivity check, not an exposure-matched causal comparison.

Table 4.7: Tunnel and random-trigger sensitivity summary

Result compared between tunnel and random triggers	Checkpoints agreeing in direction	Interpretation
Stale-attention shift	10/10	point-estimate directions agree for every checkpoint
Probability-DEF shift	10/10	checkpoint-specific directions agree, including decreases
Exposure rate	0/10 matched	random condition has much lower realized exposure

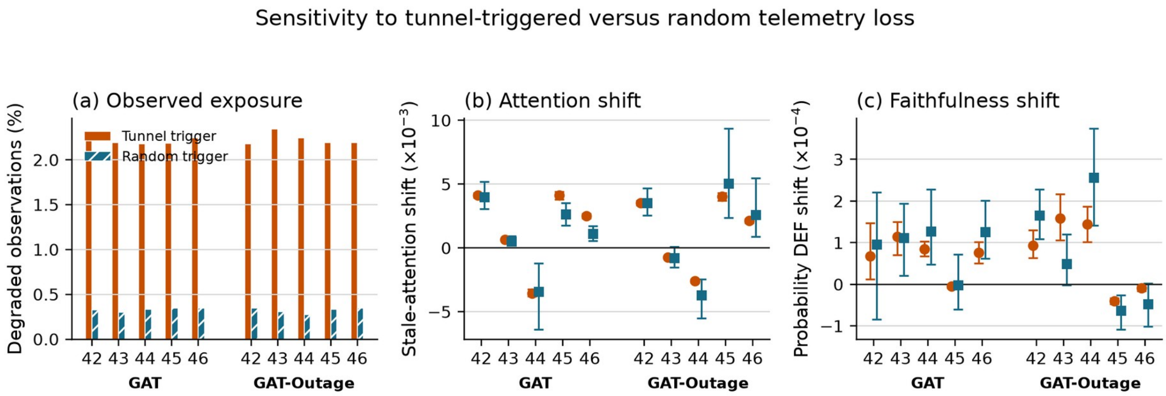

Figure 4.10. Tunnel and random triggers under a 30-second freeze. The random condition creates less exposure. Point-estimate directions agree for all ten checkpoints for both metrics, but several random-condition CIs are wide because few stale-exposed decisions occur.

4.9 Hypothesis results

Figure 4.11 reports the within-episode association between WAMSN and DEF. Seven checkpoint estimates are positive and three are negative. The negative relationship predicted by H4 is supported after Holm correction for the two seed-45 checkpoints only.

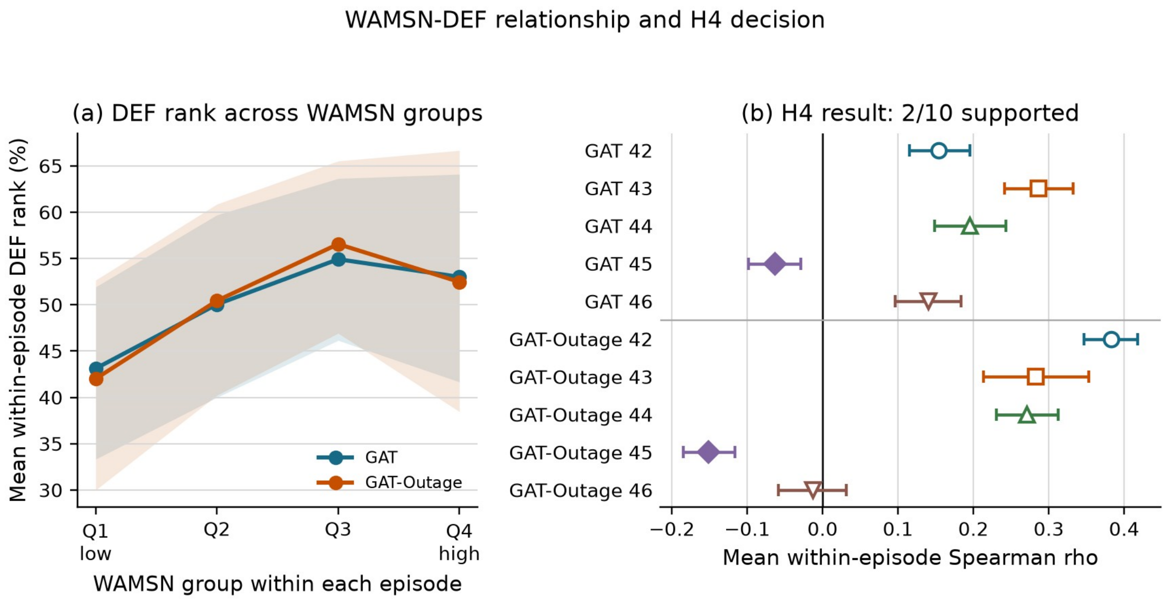

Figure 4.11. Panel (a) groups WAMSN within each episode and shows mean DEF rank. Panel (b) reports the per-checkpoint mean Spearman correlation and CI. H4 is supported for GAT and GAT-Outage seed 45, with no c onsistent support across all training seeds.

Table 4.8: Hypothesis outcomes across training seeds

Hypothesis	GAT seeds supporting	GAT-Outage seeds supporting	Verdict
H1: DEF decreases as outage duration increases	1/5	2/5	checkpoint-dependent support
H2: faithfulness declines faster than performance	1/5	1/5	limited exploratory support
H3: WAMSN increases as outage duration increases	5/5	5/5	supported manipulation check
H4: higher WAMSN is associated with lower DEF	1/5	1/5	checkpoint-dependent support
H5: GAT-Outage gives weaker H1 and H4 relationships	n/a	2/5 matched comparisons	not consistent across seeds

H1 correlations range from -0.114 to +0.198. H1 is supported for GAT seed 45 and GAT-Outage seeds 45 and 46; the latter is close to the 0.05 boundary (Holm-adjusted p=0.0495). H4 mean within-episode correlations range from -0.152 to +0.383. Both seed-45 checkpoints support H1, H2 and H4, so the proposed response occurs for some trained policies. It is not a general result across checkpoints.

H2 remains exploratory because its clean-relative rate can become large when clean DEF is close to zero; GAT-Outage seed 45 illustrates this sensitivity. GAT-Outage has weaker absolute H1 and H4 associations than its matched GAT checkpoint for seeds 43 and 46 only. H5 is descriptive and does not establish that outage training improves explanation reliability.

4.10 Answers to the research questions

RQ1: The tested averaged self-row attention does not establish reliable dispatch decision relevance under clean telemetry. Its dispatch margin-DEF is below the matched-random control, while LOO gives a positive control response. The selected request row is positive for GAT-Outage but negative for GAT; these results leave room for alternative attention-based explanation methods while showing that the extraction rule matters. No-op evidence is reported separately in Section 4.12.

RQ2: Degradation changes stale-node attention, but not in one consistent direction. Seven checkpoints show a clear shift toward stale nodes and three show a clear shift away.

RQ3: Longer outages consistently increase stale exposure, but a decline in DEF occurs for some checkpoints but is not consistent across seeds. The result also differs by action type and by attention-extraction choice.

RQ4: The tested outage-trained configuration does not remove checkpoint variation or make raw attention pass the decision-relevance check.

4.11 Explanation-release audit

The final framework decision is WITHHOLD for GAT, GAT-Outage, and all ten checkpoints. This means that the evidence is complete enough to audit, but raw attention should not be presented as the reason for a dispatch or no-op decision. This is a conservative decision for the whole declared explanation channel, based in part on failed dispatch decision relevance. It does not mean that no-op faithfulness fails individually in every checkpoint. The exploratory no-op analysis in Section 4.12 does not alter the original release criteria.

Table 4.9: Freshness-aware explanation audit decisions

Audit check	GAT	GAT-Outage	Main reason
Evidence integrity	PASS	PASS	all sweep preflights passed
Freshness visibility	PASS	PASS	AoI-derived exposure is reported separately
Action-aware controls	PASS	PASS	type matching, action protection, and LOO are present
Decision relevance	FAIL	FAIL	pooled dispatch self-row margin-DEF CIs are below zero
Action composition	PASS	PASS	no-op and dispatch are reported separately
Extraction stability	FAIL	FAIL	alternative heads or layers reverse direction
Checkpoint consistency	FAIL	FAIL	stale-attention direction differs across seeds
Deployment-action capability	NOT APPLICABLE	NOT APPLICABLE	audit scope uses sampled actions
Trigger robustness	INDETERMINATE	INDETERMINATE	intervals cross zero for GAT seeds 42/45 and GAT-Outage seeds 43/46
Model decision	WITHHOLD	WITHHOLD	required release checks do not pass

GAT seed 45 passes its individual dispatch-relevance check, but fails extraction stability and has indeterminate trigger robustness. The model-level failure must not be interpreted as failure of every individual checkpoint on every check.

Under this decision, attention remains available for internal model diagnosis, with freshness displayed separately. It is not released as an audited explanation of the selected action.

4.12 Supplementary no-op decision relevance

A supplementary analysis of the archived main-sweep records examines whether no-op attention exceeds its type-matched random control in absolute DEF. This addresses a different question from the paired change under degradation. The analysis is exploratory and leaves the original H1-H5 tests and release criteria unchanged. Only scored no-op decisions with at least one available request are eligible. The existing cadence and stale-exposure sampling schedule is retained, so these estimates describe the scored records rather than all no-op actions or a matched clean/degraded sample.

Each cell first averages scores within evaluation-seed/episode blocks for one fixed checkpoint. Table 4.10 reports equal-weight block means and pointwise 95% percentile intervals from 10,000 bootstrap draws (seed 20260912). These intervals are exploratory, are not multiplicity-adjusted, and do not estimate between-training-seed uncertainty. E/D denotes eligible episode blocks/scored decisions; episodes without eligible no-op records do not contribute.

Table 4.10: Exploratory no-op margin-DEF by checkpoint and condition

Model / seed	Clean mean [95% CI]	60 s mean [95% CI]	E/D clean; 60 s
GAT / 42	+0.498 [+0.314, +0.683]	+0.336 [+0.239, +0.425]	38/66; 48/388
GAT / 43	+0.212 [-0.008, +0.453]	-0.604 [-1.327, -0.088]	30/43; 48/288
GAT / 44	+0.567 [+0.260, +0.906]	+0.003 [-0.157, +0.171]	35/52; 48/326
GAT / 45	-0.086 [-0.189, +0.014]	+0.022 [-0.006, +0.050]	36/54; 48/342
GAT / 46	+0.888 [+0.342, +1.606]	+0.470 [+0.214, +0.760]	35/55; 48/356
GAT-Outage / 42	+0.869 [+0.680, +1.063]	+0.165 [+0.086, +0.243]	36/54; 48/343
GAT-Outage / 43	-4.049 [-8.877, -0.006]	-8.090 [-11.038, -5.530]	10/14; 37/95
GAT-Outage / 44	+0.112 [+0.029, +0.208]	-0.592 [-0.704, -0.477]	39/73; 48/412
GAT-Outage / 45	-0.023 [-0.136, +0.081]	+0.146 [+0.105, +0.185]	35/49; 48/306
GAT-Outage / 46	+0.566 [-0.021, +1.226]	+1.053 [+0.732, +1.383]	29/41; 48/276

All margin-DEF values in Table 4.10 are multiplied by 1,000 for readability. GAT seeds 42 and 46 and GAT-Outage seed 42 have positive intervals in both conditions for both margin and probability DEF. Other checkpoints have at least one negative or inconclusive interval. A decline under degradation need not imply an absolute failure against the random comparator, and the pooled dispatch result must not be generalized to every no-op explanation. Small effects, score choice, sampling differences and checkpoint variation limit interpretation.

The original whole-channel decision remains WITHHOLD because its dispatch and stability requirements are not met. This supplementary analysis does not certify any checkpoint for explanation release. Appendix C summarizes the rerun hypothesis tests and identifies the supporting files containing full-precision estimates, sample counts and archive hashes.

Chapter 5: Discussion

5.1 Answer to the central problem

The central problem is whether a graph-attention map can still be trusted when some vehicle readings are stale. In the tested setting, the averaged self-row attention channel does not meet the specified explanation-release requirements. The attention map remains available during an outage, but degradation does not have one consistent effect on faithfulness. The pooled clean dispatch control fails, while paired probability DEF has positive intervals in seven checkpoints and negative intervals in two. The evidence is insufficient to show the tested attention map as an explanation. It does not establish that stale data always cause attention explanations to fail.

This finding is about explanation reliability, not dispatch performance. A policy can still select an action while the evidence shown as its explanation is stale or weakly related to that action. Stable task output therefore cannot be used to validate the attention map.

The practical result is a release decision. Raw attention should remain an internal diagnostic unless separate tests establish freshness visibility and decision relevance. In this study, those tests do not provide sufficiently consistent evidence, so every tested checkpoint receives WITHHOLD.

5.2 How to read the evidence

The experiment provides four related but different forms of evidence.

Clean baseline. Under clean telemetry, pooled raw-attention margin-DEF is below its type-matched random control for both models, while the LOO positive control is clearly above zero. The evaluator can therefore reward a ranking built from output changes, but raw attention does not provide the required evidence of decision relevance.

Stale-data exposure. WAMSN rises as the outage becomes longer. This shows that attention-weighted vehicle-information age increases. WAMSN can rise without any attention reallocation when the same information becomes older; reallocation is assessed separately using stale-attention share and mass. It does not show that data age causes faithfulness to decline because AoI is part of the WAMSN definition.

Duration and within-policy association. The proposed negative relationships are supported for a minority of checkpoints. Both seed-45 checkpoints support H1 and H4, and GAT-Outage seed 46 additionally supports H1. The evidence therefore shows checkpoint dependence rather than a uniform decline. Conversely, a positive DEF change in other checkpoints is not sufficient to validate explanation quality.

The H1 finding for GAT-Outage seed 46 is borderline and changes under the four-decimal sensitivity check. The two seed-45 results remain supported under that check. This distinction matters when interpreting the minority of supporting checkpoints.

Direct paired change. The clean/degraded pairs compare the same decision before and after the observation-layer change. Seven checkpoints shift attention toward stale nodes and three shift away. Paired DEF intervals are positive in seven checkpoints, negative in both seed-45 checkpoints, and inconclusive for GAT-Outage seed 46. The direction cannot be generalized from one checkpoint. The pooled clean decision-relevance control also fails, so a positive paired change alone does not establish a useful explanation.

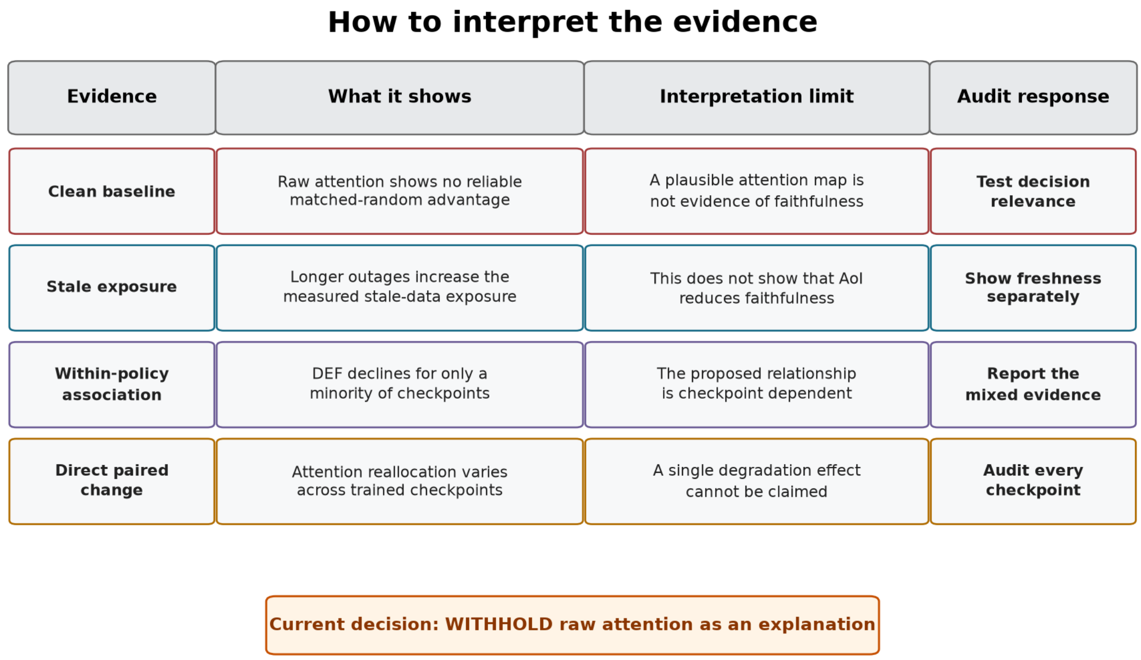

Figure 5.1. How the four forms of evidence should be interpreted. Each form answers a different question and leads to a separate audit requirement.

These measures answer different questions. WAMSN checks whether stale evidence is present, the duration tests compare progressively longer outages, and paired shifts measure the direct response of the same decision to a changed observation. Keeping them separate prevents a freshness measure from being presented as a faithfulness effect.

The negative clean result is consistent with the need to test attention rather than equate it with explanation (Jain and Wallace, 2019; Wiegreffe and Pinter, 2019). It also fits the argument of Shin et al. (2025) that naive attention aggregation can miss computation paths. It does not independently validate their GAtt method, which was not implemented here. The selected-request-row control gives positive margin-DEF for GAT-Outage in both clean and 60-second conditions, while GAT remains negative. This result supports further testing of the alternative row under a separately fixed protocol. It does not retrospectively validate the declared self-row channel, and choosing a favorable extraction rule after inspecting results would require new evaluation.

The separation between stable capability and weak explanation evidence is also relevant to Li et al. (2024) and Azzolin et al. (2026). Their settings show that correct predictions alone do not guarantee reliable explanations. The present study finds a similar problem in sequential decision-making. The mechanisms differ, however: this experiment freezes telemetry features and audits raw attention, rather than attacking graph edges or constructing self-explaining classifiers. Its evidence should not be reported as a replication of their mechanisms or as a comparable numerical effect size.

5.3 Why the result is not consistent

Checkpoint dependence. Independently trained checkpoints do not respond in the same direction. The pattern appears under both tunnel and random triggers, which reduces concern that it is produced only by the chosen tunnel. However, the random trigger does not create identical stale-node populations and cannot rule out scenario effects. The result therefore supports checkpoint dependence, not a general effect of telemetry loss.

Action composition. Dispatch accounts for 74.4% of GAT and 78.3% of GAT-Outage scorable decisions. No-op and dispatch paired DEF move in opposite directions in nine checkpoints. GAT seed 45 decreases in both groups; GAT-Outage seed 45 increases for no-op but decreases for dispatch. Reporting only the combined estimate would hide these differences.

Attention extraction. Different heads, layers, aggregations, or query rows can change the direction or strength of the result. The network does not supply a single, uniquely defined attention map. Any map shown to an operator therefore needs a predeclared extraction rule that has been tested for stability.

Action rule. The main audit samples actions from the learned policy, matching the declared scope. The smaller deterministic diagnostic also completes journeys, but it is not part of the confirmatory audit. A deployment audit must use the same action rule that the deployed system will use.

5.4 Why the construct-validity controls matter

A passenger-request node represents both information and an available action. Removing it can therefore remove the action being explained. A peer-taxi node is different: removing it hides vehicle information but leaves the action space unchanged. A uniform random baseline would mix these two effects and could make an explanation appear stronger simply because an action disappeared.

The type-matched, action-protected baseline makes the comparison fairer. It compares request nodes with request nodes and taxi nodes with taxi nodes, while protecting the selected request action. A positive DEF can then be interpreted as an advantage over comparable random information rather than an advantage created by deleting the chosen action.

LOO serves as a positive control. It ranks each node by the output change caused by removing that node, and it produces a clearer DEF response than raw attention for every checkpoint. This shows that the evaluator can respond to a more informative perturbation ranking. LOO is not a ground-truth explanation, and it does not prove that the evaluator captures every form of decision relevance.

The overlap check adds one final limit. In a small graph, explanation and random top-k sets increasingly share the same nodes as k grows. This reduces the resolution of DEF. Within this limited resolution, the tested attention channel lacks reproducible evidence of faithfulness. Other extraction or attribution methods would require their own evaluation.

The controls address a specific instance of the broader evaluator problem identified by Zheng et al. (2024): perturbations can change what the score measures. Type matching controls node composition and selected-action protection prevents deletion of the explained action. Neither guarantees that every masked graph lies on the training distribution. Similarly, Azzolin et al. (2025) show why different definitions of faithfulness should not be treated as equivalent. Here probability DEF, margin-DEF and WAMSN have explicitly different roles. He et al. (2025) further caution, through their theoretical connections, that gradient and perturbation evidence may share mechanisms. LOO therefore shows that the evaluator responds to the ranking; it does not independently establish a causal explanation.

5.5 Why outage training did not solve the problem

The tested configuration trained with 30-second outages does not solve the explanation problem. GAT-Outage does not make the direct clean/degraded response more consistent across independently trained policies. Training with stale observations may improve robustness for other purposes, but the present results do not show that it makes raw attention a more reliable explanation.

Training budgets and optimization settings are matched, but the validation telemetry differs between conditions. This prevents isolation of training telemetry alone. The reward also contains no term for explanation stability or faithfulness. A future test of model improvement should hold the selection environment fixed and assess an explicit explanation objective on new data.

Recent delay-aware methods such as DFBT (Wu et al., 2025) and robust communication methods such as MAGI (Ding et al., 2024) optimize state estimation or useful information exchange. They suggest candidate model improvements, but their objectives differ from the present release criteria. A future comparison would have to evaluate dispatch capability and apply the same action-aware explanation protocol without changing its rules. Cross-paper rewards or prediction accuracies cannot substitute for this matched comparison because tasks, action spaces and degradation mechanisms differ.

5.6 Proposed audit framework

5.6.1 Purpose and scope

The **freshness-aware explanation audit framework** combines these checks to decide whether there is enough evidence to show an attention map as an explanation to an operator. It evaluates the evidence for a frozen policy within specified operating conditions. Table 5.1 links each observed risk to the required evidence and decision rule.

Table 5.1: Freshness-aware explanation audit framework

Observed risk	Required check	Evidence	Decision rule
Attention may be attached to stale data	Report freshness separately	AoI and WAMSN	Never infer freshness from attention strength
Attention may not identify decision-relevant nodes	Use action-aware, type-matched tests	DEF, margin-DEF, and LOO	Do not claim a reason without evidence of decision relevance
Combined results may be dominated by the majority action group	Report no-op and dispatch groups	Action counts and action-specific DEF	Do not generalize a combined result across action types
The result may depend on attention extraction	Test the query row, layer, head, and aggregation	Sensitivity audits	Predeclare the extraction rule and report material sensitivity
One checkpoint may not represent another	Audit each checkpoint before release	Per-checkpoint results	Repeat after retraining or replacement
A pattern may not reproduce	Compare independent training runs	Results across seeds	Require a consistent conclusion across runs
The audited action may differ from deployment	Test the intended deployment action rule	Sampled-policy or argmax capability evidence	Require argmax capability only when deployment uses argmax

Observed risk	Required check	Evidence	Decision rule
A tunnel-specific pattern may not transfer	Compare tunnel and random triggers	Paired shifts and 95% confidence intervals	Withhold when supported directions conflict or remain indeterminate

The framework is intended for offline release review before an attention map is shown to a dispatcher or other operator. It should be repeated after retraining, checkpoint replacement, changes to the telemetry or graph pipeline, transfer to a new operating scenario, and during periodic model review. It is not a live freshness monitor. A deployed system must still display freshness separately and suppress or warn about explanations based on stale inputs. Figure 5.2 shows the corresponding offline audit workflow.

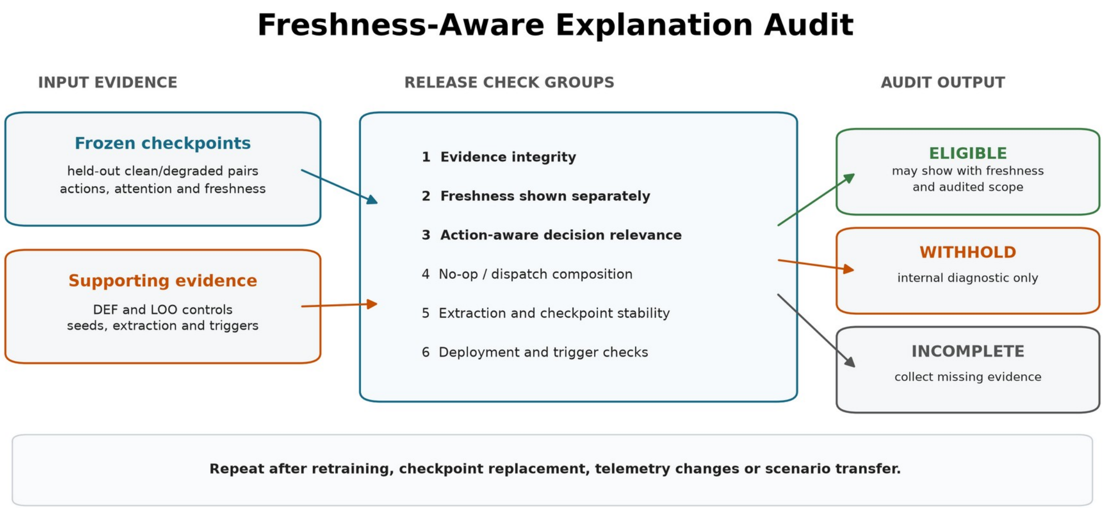

Figure 5.2. Saved evidence from frozen checkpoints passes through nine release checks, grouped here into six categories. The output controls whether attention may be presented as an explanation; it does not alter the model or selected action.

5.6.2 Inputs, outputs and use

The framework accepts evidence rather than raw attention alone. Table 5.2 summarizes its inputs and outputs.

Table 5.2: Audit framework inputs and outputs

Component	Contents	Role in the audit
Candidate scope	frozen checkpoints, model names, training seeds, and held-out conditions	defines exactly what the decision covers
Freshness evidence	paired clean/degraded records, AoI, stale-node masks, WAMSN, and stale-attention shift	separates data age from attention strength
Faithfulness evidence	type-matched and action-protected DEF, LOO, and dispatch-action results	tests whether highlighted nodes matter to the action
Stability evidence	action groups, extraction alternatives, checkpoint synthesis, deployment-action diagnostic, and trigger comparison	tests whether the conclusion survives relevant choices
Audit output	per-check status, per-checkpoint decision, model decision, failed reasons, and permitted use	supports a traceable explanation-release decision

The operating procedure is:

1. freeze the candidate checkpoint and evaluation protocol;
2. generate paired freshness, action-aware faithfulness, action-composition, and sensitivity evidence on held-out demand;
3. run preflight to establish that the evidence is complete and valid to analyze;
4. apply every release check to each checkpoint and then compare independent training runs;
5. present attention externally only if the result is ELIGIBLE, together with its freshness indicator and audited scope.

A preflight PASS is not an explanation-release pass. It only confirms that the evidence can be analyzed. The release output is ELIGIBLE, WITHHOLD, or INCOMPLETE. Individual checks may also be INDETERMINATE when the evidence is complete but does not support a direction. A failed or indeterminate required check produces WITHHOLD; missing evidence produces INCOMPLETE. In both cases, attention must not be presented as the reason for the action.

5.6.3 Application to this study

Applying the framework to the completed evidence gives WITHHOLD for GAT and GAT-Outage and for all ten individual checkpoints. The decision follows from insufficient support for explanation release: the dispatch-action decision-relevance intervals do not exceed the matched-random boundary, attention extraction can reverse the stale-attention direction, and the stale-attention response changes direction across training seeds. The random-trigger check is indeterminate for both models: intervals cross zero for GAT seeds 42/45 and GAT-Outage seeds 43/46, despite agreement in point-estimate directions. GAT seed 45 passes its individual dispatch-relevance check, but still fails extraction stability. The argmax diagnostic completes journeys, but it remains descriptive because the audit scope uses sampled actions.

The supplementary no-op results (Section 4.12) include positive absolute margin-DEF for some checkpoints in both clean and degraded conditions. The whole-channel WITHHOLD decision should therefore not be read as a universal no-op failure. H5 also describes weaker associations rather than reduced variation; consistency is assessed from per-checkpoint results.

Sampled dispatch performance provides evidence of task capability, but it cannot show that the displayed reason is current or decision-relevant.

The framework checks whether an explanation has enough evidence to be shown; it does not make the model itself more faithful. Improving the model would require an explicit explanation objective and independent evaluation, as discussed in Section 5.8.

The current study demonstrates the framework's rejection path but does not show that a real trained model can reach ELIGIBLE. The LOO positive control shows that the evaluator can reward a more informative ranking, while software tests confirm that complete passing evidence produces an ELIGIBLE decision. A future model trained for faithfulness is still needed to demonstrate that a trained policy can meet all the release requirements. Future evaluations should retain criteria fixed before testing. The audit's confidence intervals also differ from the formal model-level sufficiency certificates of Saha and Bandyopadhyay (2026).

The present framework provides release checks for the stated operating conditions, but it offers no equivalent distribution-free guarantee.

5.7 Limitations

The study uses one simulated district, 20 taxis, and 50 requests. Tunnel exposure remains scenario-dependent, although the event-aware audit scores every observed stale-exposed decision. It evaluates five independent training seeds per model. Five seeds reveal variation but provide limited information about the wider distribution of training outcomes. The selected policies pass the training stability gates, but their held-out completed-journey results overlap the legal-random CI and the simple greedy range. These baselines show that the policies can complete journeys, but do not establish a performance advantage. The experiment also does not isolate the contribution of graph edges to policy capability.

The audit therefore concerns trained graph policies whose performance has not been shown to depend on their learned action ranking or graph structure. This limits transfer to stronger policies, but does not remove the need to test whether the displayed attention reflects the audited decisions. Similar completion means do not prove that the policies behave identically or that the environment determines the outcome. The shared reward may make individual credit assignment difficult, but the coefficients in Equations 3.3 and 3.4 alone cannot establish this mechanism. Reward-component analysis and graph or action-ranking ablations would be needed to test it.

GAT and GAT-Outage share a 50-epoch budget, optimizer settings and checkpoint-selection criterion. However, their validation observations match their respective training conditions. The matched-seed comparison therefore describes two training-and-selection configurations and cannot isolate training telemetry alone. The protocol was refined during pilot work on the same scenario collection; the rerun does not provide independent validation on a previously unexamined dataset.

The local graph is small. Type matching makes random top-k sets overlap heavily with the explanation, which compresses DEF. LOO is based on the same single-node perturbation as comprehensiveness and is not an independent ground truth explanation. Other perturbation choices or post-hoc explainers may give different results.

The degradation layer freezes position and speed for fixed windows. Real telemetry may include delayed packets, partial failures, map-matching errors, and readings that resume updating at different times. The study audits attention as an explanation; it does not claim that attention is useless inside the policy computation.

The paired clean/degraded comparison supports internal validity for the tested observation-layer change because both observations come from the same simulated traffic state. Simulation and scenario-design bias remain. Road-network choice, generated demand, vehicle behavior, and tunnel placement may affect which taxis become stale and how those taxis are positioned in the local graph. In particular, taxis exposed at a fixed tunnel may differ from those elsewhere if traffic conditions at the tunnel differ from the rest of the network. The conclusions are therefore limited to the tested SUMO setting and should not be read as population estimates for real fleets or cities.

The descriptive EDA adds another boundary: test demand has a lower median straight-line origin-destination separation than training demand, while all splits share taxi initialization, background traffic and road geometry. Repeating evaluation with different seeds varies the sampled actions; it does not test new cities or independently generated demand processes. The full rerun records the workstation configuration and execution timings, but these measurements describe one machine and workload. They do not establish computational scalability or a hardware speed advantage.

5.8 Future work

Three directions follow from these limitations. The first is a comparison that holds validation telemetry fixed as well as training budgets and optimization settings, with more independent training seeds. Capability and paired DEF should be evaluated on newly reserved demand data. Measuring peak memory and computational cost across larger fleets would extend the workstation timings reported here.

The second is broader evaluation across road networks, demand processes, public mobility traces and telemetry failures. Varying tunnel placement and calibrating random-loss rates on development data would help separate trigger location, exposure frequency and outage duration. Delayed, intermittent and biased telemetry should be compared with fixed freezes, with realized exposure reported alongside faithfulness. Disconnected-edge and message-passing ablations would clarify the contribution of the graph to dispatch capability.

The third is to evaluate alternative graph explainers and training objectives that reward explanation consistency. Computation-path attribution and graph-specific post-hoc methods could be compared with LOO on independent checkpoints using release criteria fixed before testing. Both rejected and eligible candidates are needed to evaluate the framework fully. A subsequent human-factors study could assess whether freshness warnings help operators interpret the explanations; COViz (Amitai, Septon and Amir, 2024) illustrates why human understanding requires a separate evaluation.

Chapter 6: Conclusion

This dissertation asked whether raw graph-attention weights can be trusted as explanations of fleet-dispatch decisions when vehicle telemetry becomes stale. The tested weights should not be released as explanations. They remain available under clean and degraded observations, but they do not pass the required checks for decision relevance, extraction stability, and consistency across independently trained checkpoints. This conclusion concerns the declared averaged self-row channel, not all attention-based explanation methods.

Longer outages consistently increase stale-data exposure, confirming that the manipulation worked. A decline in DEF is supported for a minority of checkpoints: H1 is supported for three of ten checkpoints and H4 for two. Both seed-45 checkpoints support H1, H2 and H4. Paired DEF intervals are positive in seven checkpoints, negative in two, and inconclusive in one. These mixed results do not establish a uniform effect of degradation, and positive shifts do not by themselves validate explanation quality. Telemetry freshness and explanation faithfulness require separate checks.

The research questions are answered as follows.

1. **RQ1:** The tested averaged self-row attention does not establish reliable dispatch decision relevance under clean telemetry. Dispatch margin-DEF is below the matched-random control for both models, while LOO gives a positive control response. The selected request row is positive for GAT-Outage but negative for GAT, and supplementary no-op results vary by checkpoint and condition. These findings do not exclude other attention-based explanation methods.
2. **RQ2:** Degradation changes the attention assigned to stale nodes, but the direction is not consistent. Seven checkpoints shift attention toward stale nodes and three shift away.
3. **RQ3:** Outage duration consistently increases WAMSN, but the proposed decline in DEF is supported only for some checkpoints. The measured response also differs between no-op and dispatch actions and can reverse under alternative attention-extraction choices.
4. **RQ4:** The tested GAT-Outage configuration does not make raw attention reliably decision-relevant or remove checkpoint variation. Although training budgets and optimization settings are matched, validation telemetry differs between the two configurations, so this comparison does not isolate training telemetry alone.

The construct-validity audit is essential to these answers. A passenger-request node represents both information and an available action, whereas a peer-taxi node represents information only. Type matching and chosen-action protection prevent request deletion from creating an artificial faithfulness result. The positive LOO response shows that the evaluator can reward a ranking based on output sensitivity. The top-k overlap analysis also shows that the small graph limits measurement resolution.

The main practical contribution is the freshness-aware explanation audit framework. It accepts frozen checkpoints and held-out evidence, checks freshness, decision relevance, action composition, extraction stability, checkpoint consistency, action-rule capability, and trigger sensitivity, and returns ELIGIBLE, WITHHOLD, or INCOMPLETE. In this study, both model families and all ten checkpoints receive WITHHOLD. The evidence is complete, but it does not justify presenting attention as the reason for a dispatch or no-op decision under the declared whole-channel release rule. This conservative rule does not assert that every individual no-op explanation is unfaithful.

The study is limited to one simulated district, five training seeds per model, three held-out demand variants, different validation telemetry conditions and a small action-linked graph. The experiment demonstrates rejection of the tested raw-attention channel; it does not yet demonstrate a trained model that meets all the release requirements. It also does not measure human understanding or generalization to real fleet telemetry.

Future work should hold validation telemetry fixed when comparing clean and outage training, use more independent seeds and newly reserved demand, and extend evaluation to other road networks and calibrated telemetry-loss conditions. Explanation-aware training and alternative graph attribution methods offer further candidates for evaluation. These experiments should retain fixed release criteria, report both passing and failing checkpoints, and record hardware and timing details.

Any future release decision should remain tied to its evaluated conditions, with telemetry freshness shown separately.

List of Publications

No publication is claimed in this dissertation.

References

Abnar, S. and Zuidema, W. (2020) 'Quantifying attention flow in transformers', in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4190-4197. Available at: https://doi.org/10.18653/v1/2020.acl-main.385.

Adebayo, J. et al. (2018) 'Sanity checks for saliency maps', *Advances in Neural Information Processing Systems*, vol. 31. Available at: https://proceedings.neurips.cc/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html.

Amara, K. et al. (2022) 'GraphFramEx: Towards systematic evaluation of explainability methods for graph neural networks', in *Proceedings of the First Learning on Graphs Conference*, *Proceedings of Machine Learning Research*, vol. 198, pp. 44:1-44:23. Available at: https://proceedings.mlr.press/v198/amara22a.html.

Amitai, Y., Septon, Y. and Amir, O. (2024) 'Explaining reinforcement learning agents through counterfactual action outcomes', *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 9, pp. 10003-10011. Available at: https://doi.org/10.1609/aaai.v38i9.28863.

Ancona, M., Ceolini, E., Öztireli, C. and Gross, M. (2018) 'Towards better understanding of gradient-based attribution methods for deep neural networks', in *Proceedings of the International Conference on Learning Representations*. Available at: https://openreview.net/forum?id=Sy21R9JAW.

Armgaan, B., Dalmia, M., Medya, S. and Ranu, S. (2024) 'GraphTrail: Translating GNN predictions into human-interpretable logical rules', *Advances in Neural Information Processing Systems*, vol. 37. Available at: https://proceedings.neurips.cc/paper_files/paper/2024/hash/df2d51e1d3e899241c5c4c779c1d509f-Abstract-Conference.html.

Azzolin, S., Longa, A., Teso, S. and Passerini, A. (2025) 'Reconsidering faithfulness in regular, self-explainable and domain invariant GNNs', in *Proceedings of the 13th International Conference on Learning Representations*. Available at: https://proceedings.iclr.cc/paper_files/paper/2025/hash/7969d456aede0ce6b36fb47cc64c495c-Abstract-Conference.html.

Azzolin, S., Teso, S., Lepri, B., Passerini, A. and Malhotra, S. (2026) 'GNN explanations that do not explain and how to find them', in *Proceedings of the 14th International Conference on Learning Representations*. Available at: https://proceedings.iclr.cc/paper_files/paper/2026/hash/75a58cfb4d0de35db7663714e0f52dfa-Abstract-Conference.html.

Bekkemoen, Y. (2024) 'Explainable reinforcement learning (XRL): A systematic literature review and taxonomy', *Machine Learning*, vol. 113, no. 1, pp. 355-441. Available at: https://doi.org/10.1007/s10994-023-06479-7.

Bouteiller, Y., Ramstedt, S., Beltrame, G., Pal, C.J. and Binas, J. (2021) 'Reinforcement learning with random delays', in *Proceedings of the 9th International Conference on Learning Representations*. Available at: https://openreview.net/forum?id=QFYnKlBJYR.

Bui, N., Nguyen, H.T., Nguyen, V.A. and Ying, R. (2024) 'Explaining graph neural networks via structure-aware interaction index', in *Proceedings of the 41st International Conference on Machine Learning, PMLR*, vol. 235, pp. 4854-4883. Available at: https://proceedings.mlr.press/v235/bui24b.html.

DeYoung, J. et al. (2020) 'ERASER: A benchmark to evaluate rationalized NLP models', in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4443-4458. Available at: https://doi.org/10.18653/v1/2020.acl-main.408.

Ding, S., Du, W., Ding, L., Guo, L. and Zhang, J. (2024) 'Learning efficient and robust multi-agent communication via graph information bottleneck', *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 16, pp. 17346-17353. Available at: https://doi.org/10.1609/aaai.v38i16.29682.

Greydanus, S. et al. (2018) 'Visualizing and understanding Atari agents', in *Proceedings of the 35th International Conference on Machine Learning*, *Proceedings of Machine Learning Research*, vol. 80, pp. 1792-1801. Available at: https://proceedings.mlr.press/v80/greydanus18a.html.

He, J., Rafiey, A., Mishne, G. and Wang, Y. (2025) 'Explaining GNN explanations with edge gradients', in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, vol. 2, pp. 884-895. Available at: https://doi.org/10.1145/3711896.3736947.

Heuillet, A., Couthouis, F. and Díaz-Rodríguez, N. (2021) 'Explainability in deep reinforcement learning', *Knowledge-Based Systems*, vol. 214, article 106685. Available at: https://doi.org/10.1016/j.knosys.2020.106685.

Hong, S., Liu, Y., Li, Z., Li, S. and He, Y. (2024) 'Multi-agent collaborative perception via motion-aware robust communication network', in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15301-15310. Available at: https://openaccess.thecvf.com/content/CVPR2024/html/Hong_Multi-agent_Collaborative_Perception_via_Motion-aware_Robust_Communication_Network_CVPR_2024_paper.html.

Hooker, S. et al. (2019) 'A benchmark for interpretability methods in deep neural networks', *Advances in Neural Information Processing Systems*, vol. 32. Available at: https://proceedings.neurips.cc/paper/2019/hash/fe4b8556000d0f0cae99daa5c5c5a410-Abstract.html.

Hu, Y., Feng, S. and Li, S. (2025) 'BMG-Q: Localized bipartite match graph attention Q-learning for ride-pooling order dispatch', *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 10, pp. 15407-15421. Available at: https://doi.org/10.1109/TITS.2025.3595653.

Iqbal, S. and Sha, F. (2019) 'Actor-attention-critic for multi-agent reinforcement learning', in *Proceedings of the 36th International Conference on Machine Learning*, *Proceedings of Machine Learning Research*, vol. 97, pp. 2961-2970. Available at: https://proceedings.mlr.press/v97/iqbal19a.html.

Jacovi, A. and Goldberg, Y. (2020) 'Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness?', in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4198-4205. Available at: https://doi.org/10.18653/v1/2020.acl-main.386.

Jain, S. and Wallace, B.C. (2019) 'Attention is not explanation', in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, vol. 1, pp. 3543-3556. Available at: https://aclanthology.org/N19-1357/.

Kaul, S., Yates, R. and Gruteser, M. (2012) 'Real-time status: How often should one update?', in *Proceedings of IEEE INFOCOM*, pp. 2731-2735. Available at: https://doi.org/10.1109/INFCOM.2012.6195689.

Li, J., Pang, M., Dong, Y., Jia, J. and Wang, B. (2024) 'Graph neural network explanations are fragile', in *Proceedings of the 41st International Conference on Machine Learning, Proceedings of Machine Learning Research*, vol. 235, pp. 28551-28567. Available at: https://proceedings.mlr.press/v235/li24bd.html.

Lin, K. et al. (2018) 'Efficient large-scale fleet management via multi-agent deep reinforcement learning', in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1774-1783. Available at: https://doi.org/10.1145/3219819.3219993.

Liotet, P., Maran, D., Bisi, L. and Restelli, M. (2022) 'Delayed reinforcement learning by imitation', in *Proceedings of the 39th International Conference on Machine Learning, Proceedings of Machine Learning Research*, vol. 162, pp. 13528-13556. Available at: https://proceedings.mlr.press/v162/liotet22a.html.

Lipton, Z.C. (2018) 'The mythos of model interpretability', *Queue*, vol. 16, no. 3, pp. 31-57. Available at: https://doi.org/10.1145/3236386.3241340.

Liu, Y. and Xie, S. (2025) 'Explanations of GNN on evolving graphs via axiomatic layer edges', in *Proceedings of the 13th International Conference on Learning Representations*. Available at: https://proceedings.iclr.cc/paper_files/paper/2025/hash/de88d78c875d93a2098dc49a0d0422da-Abstract-Conference.html.

Liu, Y. et al. (2022) 'Rethinking attention-model explainability through faithfulness violation test', in *Proceedings of the 39th International Conference on Machine Learning, Proceedings of Machine Learning Research*, vol. 162, pp. 13807-13824. Available at: https://proceedings.mlr.press/v162/liu22i.html.

Liu, Y., Zhang, X., Xie, S. and Xiong, H. (2025) 'Robust explanations of graph neural networks via graph curvatures', *Advances in Neural Information Processing Systems*, vol. 38. Available at: https://proceedings.neurips.cc/paper_files/paper/2025/hash/2c2e95b75a10adbd2359f8ed5c0a38cd-Abstract-Conference.html.

Lopez, P.A. et al. (2018) 'Microscopic traffic simulation using SUMO', in *Proceedings of the 21st IEEE International Conference on Intelligent Transportation Systems*, pp. 2575-2582. Available at: https://doi.org/10.1109/ITSC.2018.8569938.

Lowe, R. et al. (2017) 'Multi-agent actor-critic for mixed cooperative-competitive environments', *Advances in Neural Information Processing Systems*, vol. 30. Available at: https://proceedings.neurips.cc/paper/2017/hash/68a9750337a418a86fe06c1991a1d64c-Abstract.html.

Lu, S., Mills, K.G., He, J., Liu, B. and Niu, D. (2024) 'GOAt: Explaining graph neural networks via graph output attribution', in *Proceedings of the 12th International Conference on Learning Representations*. Available at: https://proceedings.iclr.cc/paper_files/paper/2024/hash/92ee07d8a2c8f5ec08eff83f9eff0c1b-Abstract-Conference.html.

Lu, W., Zhao, X., Fryen, T., Lee, J.H., Li, M., Magg, S. and Wermter, S. (2024) 'Causal state distillation for explainable reinforcement learning', in *Proceedings of the Third Conference on Causal Learning and Reasoning*, *PMLR*, vol. 236, pp. 106-142. Available at: https://proceedings.mlr.press/v236/lu24a.html.

Luo, D. et al. (2020) 'Parameterized explainer for graph neural network', *Advances in Neural Information Processing Systems*, vol. 33, pp. 19620-19631. Available at: https://proceedings.neurips.cc/paper/2020/hash/e37b08dd3015330dcbb5d6663667b8b8-Abstract.html.

Madumal, P. et al. (2020) 'Explainable reinforcement learning through a causal lens', *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 3, pp. 2493-2500. Available at: https://doi.org/10.1609/aaai.v34i03.5631.

Milani, S. et al. (2024) 'Explainable reinforcement learning: A survey and comparative review', *ACM Computing Surveys*, vol. 56, no. 7, pp. 1-36. Available at: https://doi.org/10.1145/3616864.

Mott, A. et al. (2019) 'Towards interpretable reinforcement learning using attention augmented agents', *Advances in Neural Information Processing Systems*, vol. 32. Available at: https://proceedings.neurips.cc/paper/2019/hash/e9510081ac30ffa83f10b68cde1cac07-Abstract.html.

OpenStreetMap contributors (n.d.) *Copyright and license*. Available at: https://www.openstreetmap.org/copyright (Accessed: 12 September 2026).

Pan, B. et al. (2025) 'GraphNarrator: Generating textual explanations for graph neural networks', in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, vol. 1, pp. 23-42. Available at: https://doi.org/10.18653/v1/2025.acl-long.2.

Qin, Z., Zhu, H. and Ye, J. (2022) 'Reinforcement learning for ridesharing: An extended survey', *Transportation Research Part C: Emerging Technologies*, vol. 144, article 103852. Available at: https://doi.org/10.1016/j.trc.2022.103852.

Rashid, T. et al. (2020) 'Monotonic value function factorisation for deep multi-agent reinforcement learning', *Journal of Machine Learning Research*, vol. 21, no. 178, pp. 1-51. Available at: https://jmlr.org/papers/v21/20-081.html.

Ribeiro, M.T., Singh, S. and Guestrin, C. (2016) 'Why should I trust you? Explaining the predictions of any classifier', in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135-1144. Available at: https://doi.org/10.1145/2939672.2939778.

Saha, S. and Bandyopadhyay, S. (2026) 'Certified evaluation of model-level explanations for graph neural networks', in *Proceedings of the 14th International Conference on Learning Representations*. Available at: https://proceedings.iclr.cc/paper_files/paper/2026/hash/c354d2a64a71776538d6dbbc38285216-Abstract-Conference.html.

Scarselli, F. et al. (2009) 'The graph neural network model', *IEEE Transactions on Neural Networks*, vol. 20, no. 1, pp. 61-80. Available at: https://doi.org/10.1109/TNN.2008.2005605.

Serrano, S. and Smith, N.A. (2019) 'Is attention interpretable?', in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 2931-2951. Available at: https://doi.org/10.18653/v1/P19-1282.

Sha, J. et al. (2026) 'A multi-agent reinforcement learning scheduling algorithm integrating state graph and task graph structural modeling for ride-sharing dispatching', *Scientific Reports*, vol. 16, article 5461. Available at: https://doi.org/10.1038/s41598-026-35004-8.

Shin, Y.-M., Li, S., Cao, X. and Shin, W.-Y. (2025) 'Faithful and accurate self-attention attribution for message passing neural networks via the computation tree viewpoint', *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 19, pp. 20461-20469. Available at: https://doi.org/10.1609/aaai.v39i19.34254.

Soudijani, S. and Dimitrova, R. (2025) 'Synthesis of communication policies for multi-agent systems robust to communication restrictions', in *Proceedings of the 34th International Joint Conference on Artificial Intelligence*, pp. 257-266. Available at: https://doi.org/10.24963/ijcai.2025/30.

Tabassi, E. (2023) *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1. Gaithersburg, MD: National Institute of Standards and Technology. Available at: https://doi.org/10.6028/NIST.AI.100-1.

Varbella, A., Amara, K., Gjorgiev, B., El-Assady, M. and Sansavini, G. (2024) 'PowerGraph: A power grid benchmark dataset for graph neural networks', *Advances in Neural Information Processing Systems*, vol. 37. Available at: https://proceedings.neurips.cc/paper_files/paper/2024/hash/c7caf017cbbca1f4b368ffdc7bb8f319-Abstract-Datasets_and_Benchmarks_Track.html.

Vaswani, A. et al. (2017) 'Attention is all you need', *Advances in Neural Information Processing Systems*, vol. 30, pp. 5998-6008. Available at: https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

Veličković, P. et al. (2018) 'Graph attention networks', in *Proceedings of the International Conference on Learning Representations*. Available at: https://openreview.net/forum?id=rJXMpikCZ.

Wang, J. et al. (2025) 'CoopRide: Cooperate all grids in city-scale ride-hailing dispatching with multi-agent reinforcement learning', in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, vol. 1, pp. 1457-1468. Available at: https://doi.org/10.1145/3690624.3709205.

Wang, X. and Shen, H.W. (2024) 'GNNBoundary: Towards explaining graph neural networks through the lens of decision boundaries', in *Proceedings of the 12th International Conference on Learning Representations*. Available at: https://proceedings.iclr.cc/paper_files/paper/2024/hash/f72d4fdfd5eb425cd81df9fe6272a533-Abstract-Conference.html.

Wiegreffe, S. and Pinter, Y. (2019) 'Attention is not not explanation', in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pp. 11-20. Available at: https://doi.org/10.18653/v1/D19-1002.

Wu, Q. et al. (2025) 'Directly forecasting belief for reinforcement learning with delays', in *Proceedings of the 42nd International Conference on Machine Learning*, *PMLR*, vol. 267, pp. 67810-67832. Available at: https://proceedings.mlr.press/v267/wu25ag.html.

Yao, Z., Florescu, I. and Lee, C. (2024) 'Control in stochastic environment with delays: A model-based reinforcement learning approach', *Proceedings of the International Conference on Automated Planning and Scheduling*, vol. 34, no. 1, pp. 663-670. Available at: https://doi.org/10.1609/icaps.v34i1.31529.

Yates, R.D. et al. (2021) 'Age of information: An introduction and survey', *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 5, pp. 1183-1210. Available at: https://doi.org/10.1109/JSAC.2021.3065072.

Ying, Z. et al. (2019) 'GNNExplainer: Generating explanations for graph neural networks', *Advances in Neural Information Processing Systems*, vol. 32, pp. 9240-9251. Available at: https://proceedings.neurips.cc/paper/2019/hash/d80b7040b773199015de6d3b4293c8ff-Abstract.html.

Yu, C. et al. (2022) 'The surprising effectiveness of PPO in cooperative multi-agent games', *Advances in Neural Information Processing Systems*, vol. 35, pp. 24611-24624. Available at: https://proceedings.neurips.cc/paper_files/paper/2022/hash/9c1535a02f0ce079433344e14d910597-Abstract-Datasets_and_Benchmarks.html.

Yu, Z. and Gao, H. (2025) 'MAGE: Model-level graph neural networks explanations via motif-based graph generation', in *Proceedings of the 13th International Conference on Learning Representations*. Available at: https://proceedings.iclr.cc/paper_files/paper/2025/hash/7749f9c0d5ff109231be21e910a3ced2-Abstract-Conference.html.

Yuan, H. et al. (2021) 'On explainability of graph neural networks via subgraph explorations', in *Proceedings of the 38th International Conference on Machine Learning*, *Proceedings of Machine Learning Research*, vol. 139, pp. 12241-12252. Available at: https://proceedings.mlr.press/v139/yuan21c.html.

Yuan, H. et al. (2023) 'Explainability in graph neural networks: A taxonomic survey', *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 5, pp. 5782-5799. Available at: https://doi.org/10.1109/TPAMI.2022.3204236.

Zhang, J., Chen, Z., Mei, H., Da, L., Luo, D. and Wei, H. (2024) 'RegExplainer: Generating explanations for graph neural networks in regression tasks', *Advances in Neural Information Processing Systems*, vol. 37. Available at: https://proceedings.neurips.cc/paper_files/paper/2024/hash/909f526db5127f8bd8158af32d9e313a-Abstract-Conference.html.

Zheng, X., Shirani, F., Wang, T., Cheng, W., Chen, Z., Chen, H., Wei, H. and Luo, D. (2024) 'Towards robust fidelity for evaluating explainability of graph neural networks', in *Proceedings of the 12th International Conference on Learning Representations*. Available at: https://proceedings.iclr.cc/paper_files/paper/2024/hash/34293d684b1012ed45c3274b4a7edc00-Abstract-Conference.html.

Zhou, H., Huang, A., Azizzadenesheli, K., Childers, D. and Lipton, Z. (2024) 'Timing as an action: Learning when to observe and act', in *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, *PMLR*, vol. 238, pp. 3979-3987. Available at: https://proceedings.mlr.press/v238/zhou24c.html.

Appendices

Appendix A: Experiment Reproduction

The complete procedure is provided in docs/REPRODUCE_EXPERIMENTS.md. The guide identifies the required environment, fixed configurations, commands, validation checks, and expected outputs. The final audit uses configs/experiments/dissertation_v12_full_rerun.toml, and its version-controlled results are stored under results/dissertation_v12_full_rerun/.

Appendix B: Supporting Artifacts

The repository contains the experiment configuration, source code, validation selections, preflight reports, per-seed analyses, summary tables, figure sources, selected checkpoints, and archived raw cells used in this dissertation. The release package includes a SHA-256 manifest for verification.

Literature search scope

The narrative review was updated through 12 September 2026 using targeted searches of primary proceedings and publisher records, including PMLR, ICLR, NeurIPS, AAAI, IJCAI, ACL, ACM and CVF. Search terms covered graph explanation faithfulness, attention attribution, evolving graph explanations, reinforcement learning observation delays, and robust multi-agent communication. Studies were selected for relevance to decision attribution, evaluation validity, temporal change, delayed observations or relational coordination. Foundational work was retained to define the models and measures. This search supports a focused narrative synthesis with no claim of exhaustive database coverage; studies from adjacent tasks inform methodological comparisons rather than direct fleet-dispatch performance comparisons.

Appendix C: Statistical details

Table C.1 reproduces the rerun checkpoint-specific H1-H4 statistics. H1 and H3 report Spearman rho; H2 reports the clean-relative rate difference and remains exploratory; H4 reports mean within-episode rho. N is the number of scored records for H1/H3, comparison cells for H2, and episodes for H4. B is the episode-block count where available. Holm adjustment is within each checkpoint's H1-H4 family. These are not cross-training-seed tests.

Table C.1: Rerun per-checkpoint hypothesis statistics

Model/seed	H	Statistic	Raw p	Holm p	N / B
GAT/42	H1	+0.081	0.9999	1	3988 / 240
GAT/42	H2	-0.5112	1	1	32 / n/a
GAT/42	H3	+0.4785	9.999e-05	0.0004	4836 / 240
GAT/42	H4	+0.1538	1	1	176 / 176
GAT/43	H1	+0.0993	1	1	4021 / 240
GAT/43	H2	-0.6685	1	1	32 / n/a
GAT/43	H3	+0.4763	9.999e-05	0.0004	4869 / 240
GAT/43	H4	+0.2867	1	1	176 / 176
GAT/44	H1	+0.05729	0.999	1	4033 / 240
GAT/44	H2	-0.6515	1	1	32 / n/a
GAT/44	H3	+0.4811	9.999e-05	0.0004	4881 / 240
GAT/44	H4	+0.1956	1	1	176 / 176
GAT/45	H1	-0.03854	0.007799	0.007799	4029 / 240
GAT/45	H2	+0.4516	9.999e-05	0.0004	32 / n/a
GAT/45	H3	+0.4828	9.999e-05	0.0004	4877 / 240
GAT/45	H4	-0.06298	0.0003	0.0005999	176 / 176
GAT/46	H1	+0.07597	0.9999	1	4025 / 240
GAT/46	H2	-0.5983	1	1	32 / n/a
GAT/46	H3	+0.4828	9.999e-05	0.0004	4873 / 240
GAT/46	H4	+0.1404	1	1	176 / 176
GAT-O/42	H1	+0.1978	1	1	4026 / 240
GAT-O/42	H2	-0.641	1	1	32 / n/a
GAT-O/42	H3	+0.4811	9.999e-05	0.0004	4874 / 240
GAT-O/42	H4	+0.3827	1	1	176 / 176
GAT-O/43	H1	+0.02185	0.7829	1	3997 / 240
GAT-O/43	H2	-0.527	1	1	32 / n/a
GAT-O/43	H3	+0.4861	9.999e-05	0.0004	4845 / 240
GAT-O/43	H4	+0.2829	1	1	176 / 176
GAT-O/44	H1	+0.123	1	1	4003 / 240
GAT-O/44	H2	-1.486	1	1	32 / n/a
GAT-O/44	H3	+0.4658	9.999e-05	0.0004	4851 / 240
GAT-O/44	H4	+0.2714	1	1	176 / 176
GAT-O/45	H1	-0.1137	9.999e-05	0.0004	4025 / 240
GAT-O/45	H2	+7.719	0.0002	0.0004	32 / n/a
GAT-O/45	H3	+0.481	9.999e-05	0.0004	4873 / 240

Model/seed	H	Statistic	Raw p	Holm p	N / B
GAT-O/45	H4	-0.1517	9.999e-05	0.0004	176 / 176
GAT-O/46	H1	-0.03683	0.0165	0.0495	4005 / 240
GAT-O/46	H2	-0.422	1	1	32 / n/a
GAT-O/46	H3	+0.481	9.999e-05	0.0004	4853 / 240
GAT-O/46	H4	-0.01237	0.2853	0.5705	176 / 176

GAT-O denotes GAT-Outage. Full-precision values and the H1/H3 confidence intervals are retained in hypothesis_statistics.csv under results/dissertation_v12_full_rerun/supplementary_review/. Missing intervals are not estimated retrospectively. The same directory contains all probability and margin no-op estimates, sample counts and input hashes. The completed five-seed, 50-epoch protocol and runtime are documented in docs/FULL_RERUN_V12.md.